

Model Theory and Statistical Learning Theory

Maxim Podolski

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Betreuer: Prof. Dr. Philipp Hieronymi

Zweitgutachter: Dr. Tingxiang Zou

MATHEMATISCHES INSTITUT

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RHEINISCHEN FRIEDRICH-WILHELMS-UNIVERSITÄT BONN

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Conventions

- We define $0 \in \mathbb{N}$.
- Throughout this thesis, $\log$ denotes the logarithm to base 2 ($\log_2$), unless explicitly stated otherwise.
- Given a set X , by $\mathcal{P}(X)$ we mean the power set of X .

1 Introduction

Assume that there is some $\mathcal{L}$ -structure $\mathcal{M}$, which we want to understand better by analyzing its types. Let $A \subseteq M$ and $b \in M^n$. The *type of b in $\mathcal{M}$ over A* (written: $\text{tp}^{\mathcal{M}}(b|A)$) is defined as the set of all $\mathcal{L}(A)$ -formulas $\varphi(x)$ such that $\mathcal{M} \models \varphi(b)$. Assume now that A is finite. A natural question that arises is whether we can understand types locally; if there are $a \in A$ such that $\text{tp}^{\mathcal{M}}(b|A)$ can be written as a subset of the union from these $\text{tp}^{\mathcal{M}}(a|A)$ or, more precisely, the *traces* of $\mathcal{L}$ -formulas on A can be reconstructed with local parameters but not necessarily the same formulas. The trace of a formula $\varphi(x; y)$ on a set A with respect to an element b is defined as the set $\{c \in A : \mathcal{M} \models \varphi(c; b)\}$. Informally speaking, we are looking for substitute formulas that have the same logical validity on the underlying finite set A , using parameters of A . This property is called *Uniform Definability of Types over Finite Sets* (UDTFS).

To illustrate this, consider the structure of the real numbers with their linear order, $\mathcal{M} = (\mathbb{R}, +, \cdot, \leq, 0, 1)$. Let $A = \{0, 1, 2\} \subseteq \mathbb{R}$ be a finite set and take $b = 1.5 \in \mathbb{R}$. The type $\text{tp}^{\mathcal{M}}(b|A)$ includes formulas like $0 \leq x$, $x \leq 2$, and $\neg(x \leq 1)$. Now, focus on the formula $\varphi(x; y) = x \leq y$. The trace of $\varphi(x; b)$ on A is the set $\{c \in A : c \leq 1.5\} = \{0, 1\}$. UDTFS states that for a formula $\varphi(x; y)$, there exists a formula $\vartheta(x; z)$ such that for every finite set A and every $b \in M$, there is an element $a \in A$ for which the trace of $\varphi(x; b)$ on A equals the trace of $\vartheta(x; a)$ on A . Indeed, choosing $\vartheta(x; z) := x \leq z$ and $a := 1$ works, as $\{c \in A : c \leq 1\} = \{0, 1\}$, which matches the trace of $\varphi(x; b)$ on A . This example illustrates that even for a single formula such as $x \leq y$, UDTFS allows us to *compress* the information about the trace of $\varphi(x; b)$ on A into the behavior of a single element $a \in A$ with respect to a uniform formula ϑ . This local definability is a powerful property, as it implies that the type of b over A can be understood in terms of the types of elements already in A .

However, UDTFS is a strong global property. In fact, as established in [CS12], a theory satisfies UDTFS if and only if it is NIP. This equivalence highlights that UDTFS is not just a local phenomenon but deeply tied to the combinatorial and logical structure of the theory. For theories that are not NIP UDTFS fails globally. Yet, one might wonder whether UDTFS holds locally, perhaps for certain formulas or types. This leads us to the idea of studying UDTFS in a more localized or restricted setting.

The question of whether UDTFS holds locally leads us to statistical learning theory. In this field, the study of concept classes and their VC-dimension provides a combinatorial framework that mirrors the model-theoretic properties of NIP and compression schemes. A concept class $(X, \mathcal{C})$ consists of a set X and a family $\mathcal{C}$ of subsets of X . The VC-dimension of $\mathcal{C}$ measures the complexity of the class by determining the largest set $A \subseteq X$ that can be *shattered* by $\mathcal{C}$, meaning that all subsets of A can be realized as intersections

with sets from $\mathcal{C}$. This concept is analogous to the model-theoretic notion of the VC-dimension of a formula φ , which measures the largest set A such that all subsets of A can be defined by instances of φ with parameters from A , which directly connects to the *No Independence Property* (NIP).

The interest in VC-dimensions of concept classes originates from the fact that a finite VC-dimension is equivalent to a concept class to be *PAC-learnable*, which means informally that we will most likely be able to solve the problem behind our concept class with a low amount of errors. That was proven in [Blu+89]. Now, we want to understand if the VC-dimension of a concept class can tell us even more.

A key result in learning theory is the existence of *compression schemes* for concept classes with finite VC-dimension. A compression scheme is a method to "reconstruct" a concept (an element of $\mathcal{C}$) using only a small number of examples from the training data. This is strikingly similar to UDTFS, which allows us to represent the trace of a formula $\varphi(x; b)$ on a finite set A using just a single element $a \in A$ and a uniform formula ϑ . In fact, the question of whether every concept class with finite VC-dimension has a compression scheme is a longstanding open problem in learning theory, analogous to the UDTFS conjecture in model theory.

To make this connection concrete, consider the concept class of *half-spaces* in $\mathbb{R}^d$. Here, $X = \mathbb{R}^d$ and $\mathcal{C}$ is the family of all half-spaces; sets of the form $\{x \in \mathbb{R}^d : w \cdot x + b \leq 0\}$ for some weight vector $w \in \mathbb{R}^d$ and bias $b \in \mathbb{R}$. A half-space can be viewed as the definable set of the formula $\varphi(x_1, \dots, x_d; w_1, \dots, w_d, b) := w_1 \cdot x_1 + \dots + w_d \cdot x_d + b \leq 0$ originating from our structure $\mathcal{M}$. The VC-dimension of this class is known to be $d + 1$. This means that there exists a set of $d + 1$ points in $\mathbb{R}^d$ that can be shattered by half-spaces, but no larger set can be shattered. As we will prove in this thesis, the finite VC-dimension of half-spaces guarantees an honest k -compression scheme. This means that, given a sample set of size k , we can reconstruct the underlying concept (in this case the underlying half-space), because an honest compression scheme not only compresses the concept but also ensures that the compression is consistent with the original data. This is analogous to UDTFS, where the uniform formula ϑ and the element $a \in A$ not only match the trace of $\varphi(x; b)$ on A but do so in a way that is consistent with the entire structure of the model.

Beyond its theoretical significance, bounded *honest* compression schemes provide a robust safeguard against overfitting, which typically occurs due to insufficient training data. In this context, honesty ensures that the reconstructed concept $C \in \mathcal{C}$ not only matches the target concept $C^* \in \mathcal{C}$ on the *consistent* sample set, but also satisfies $C \subseteq C^*$, eliminating false positive classifications. By showing that a concept can be at least partially reconstructed from a small sample, compression schemes ensure that the learned concept does not merely memorize local statistical inconsistencies but captures the underlying structure of the data. Crucially, the primary value of an

honest k -compression scheme lies in the guarantee it provides: it ensures that a training set of at least k consistent examples is sufficient to learn a concept that performs well on unseen data without overfitting. Thus, even without knowing the exact structure of the scheme, its mere existence guarantees that k appropriately chosen examples suffice to learn a robust predictor.

The connection between model theory and learning theory is further highlighted by the *Shelah-Sauer-Lemma*, which was first proven in model theory by Shelah in 1972 and later rediscovered in combinatorics by Sauer. This lemma bounds the number of maximal consistent sets of formulas $\varphi(x; y)$ on finite sets $A \subseteq M^{|y|}$ —where M is the universe of the structure $\mathcal{M}$ —by a polynomial in $|A|$, depending only on the VC-dimension of φ . In learning theory, this result is fundamental for bounding the sample complexity of learning algorithms.

In this thesis, we will first establish the model-theoretic background leading to the UDTFS conjecture. We will then delve into the learning-theoretic framework, proving the result of Livni and Simon that concept classes with finite VC-dimension admit honest compression schemes. Finally, we will merge these two perspectives to explore the extended UDTFS conjecture, showing how the honest compression scheme result can be used to verify UDTFS locally for certain formulas. This interdisciplinary approach not only deepens our understanding of UDTFS but also highlights the beautiful interplay between model theory and learning theory.

All contributions are marked according to the official requirements for theses in the bachelor study program Mathematik at the University of Bonn. We indicate the source material in round brackets after the theorem or lemma title. Margin notes indicate the degree of personal contribution:

- 0 indicates exact copy of the source material,
- 1 indicates minor contribution: worked out details and improved structure and clarity of the argument,
- 2 indicates major contribution: there was a significant amount of work done to extend and explain the argument,
- 3 indicates original contribution.

In some cases, we will provide a brief explanation of our exact contribution in the margin notes.

2 Meanwhile in Model Theory

We begin this thesis by recalling the fundamental definitions and results from first-order logic required for our study. After establishing these model-theoretic prerequisites, we will work our way toward the notion of Uniform Definability of Types over Finite Sets (UDTFS). When we subsequently transition to the learning-theoretic framework, we will frequently refer back to the concepts established in this chapter, as we will uncover deep functional and structural similarities between the two domains.

2.1 Prerequisites

To avoid ambiguity in the formal statements that follow, we fix our notation regarding $\mathcal{L}$ -structures, variables, and formulas.

Remark 2.1 (Notation) We work in a fixed language $\mathcal{L}$ with syntactically correct $\mathcal{L}$ -formulas.

- $\mathcal{M}$ denotes an $\mathcal{L}$ -structure with universe M . It is convenient to say that $\mathcal{M}$ is a structure when we mean that it is an $\mathcal{L}$ -structure.
- x, y, z represent tuples of variables, whose parameters are chosen from M .
- $|x|$ denotes the length of the variable tuple x .
- We will work with *partitioned formulas*, namely formulas $\varphi(x; y)$ with a separation of variables into object variables (represented by x) and parameter variables (represented by y). The intended partition is indicated by a semicolon.

Given an $\mathcal{L}$ -theory T and a model $\mathcal{M} \models T$, the central objects of study in model theory are the sets that can be carved out by first-order formulas using parameters.

Definition 2.2 (Definable set [LS13]) Let $\varphi(x; y)$ be a formula, where x and y are sequences of variables of respective lengths $|x|$ and $|y|$. Given a tuple of parameters $c \in M^{|y|}$, the set

$$\varphi(M; c) = \{a \in M^{|x|} : M \models \varphi(a, c)\} \subseteq M^{|x|}$$

is called a *definable set* of M . Furthermore, we associate with the formula $\varphi(x; y)$ the family of definable sets

$$\mathcal{F}_\varphi = \{\varphi(M; c) : c \in M^{|y|}\} \subseteq \mathcal{P}(M^{|x|}).$$

When analyzing the behavior of definable sets restricted to smaller, localized domains, we consider their traces.

Definition 2.3 (Trace of a formula [LS13]) Let $\mathcal{M}$ be an $\mathcal{L}$ -structure and $\varphi(x; y)$ an $\mathcal{L}$ -formula. If $A \subseteq M$ is an arbitrary subset of M and $b \in M^{|y|}$ a tuple of parameters, we define:

$$\varphi(A; b) = \{a \in A^{|x|} : \mathcal{M} \models \varphi(a; b)\}.$$

This set is called the *trace* of the formula $\varphi(x; b)$ on the set A .

To relate structures of different sizes while preserving their first-order expressiveness, we recall the standard notion of an elementary extension.

Definition 2.4 (Elementary extension [Hie26]) Let $\mathcal{M}$ and $\mathcal{N}$ be $\mathcal{L}$ -structures such that $\mathcal{M} \subseteq \mathcal{N}$. We say $\mathcal{N}$ is an *elementary extension* of $\mathcal{M}$ (and write $\mathcal{M} \preceq \mathcal{N}$) if for every $\mathcal{L}$ -formula $\varphi(x_1, \dots, x_n)$ and all $a_1, \dots, a_n \in M$,

$$\mathcal{M} \models \varphi(a_1, \dots, a_n) \iff \mathcal{N} \models \varphi(a_1, \dots, a_n).$$

2.2 No Independence Property

We now introduce the No Independence Property (NIP). While NIP was originally defined via indiscernible sequence (see [CS12]), it was established that NIP can be equivalently explained in terms of Vapnik-Chervonenkis VC-dimension.

Definition 2.5 (Shattering and VC-dimension [CS12]) Given a set X and a set system $\mathcal{C} \subseteq \mathcal{P}(X)$, a finite set $A \subseteq X$ is said to be *shattered* if

$$\{A \cap C : C \in \mathcal{C}\} = 2^A = \mathcal{P}(A)$$

The *Vapnik-Chervonenkis-dimension* (*VC-dimension*) of $(X, \mathcal{C})$, $\text{VC-dim}(\mathcal{C}) := \text{VC-dim}((X, \mathcal{C}))$, is defined as the maximal cardinality of a finite set A that is shattered. If such a maximum doesn't exist, we write $\text{VC-dim}(\mathcal{C}) = \infty$.

Remark 2.6 Note that in Definition 2.5, not every set $B \subseteq X$ of cardinality $|A|$ needs to be shattered.

Definition 2.7 (NIP formulas and NIP structures) A formula $\varphi(x; y)$ is said to have the *No Independence Property* or be *NIP* in short if the associated family of definable sets $\mathcal{F}_\varphi$ has finite VC-dimension as in Definition 2.5. So some formula $\varphi(x; y)$ creates the set system $\mathcal{F}_\varphi$. Now by saying $\mathcal{F}_\varphi$ has finite VC-dimension, we mean that $(M^{|x|}, \mathcal{F}_\varphi)$ has finite VC-dimension. An $\mathcal{L}$ -structure $\mathcal{M}$ is called NIP if every $\mathcal{L}$ -formula $\varphi(x; y)$ is NIP in $\mathcal{M}$. We say an $\mathcal{L}$ -theory T is NIP, if every model of T is NIP.

We denote the VC-dimension of a formula in a model $\mathcal{M}$ as

$$\text{VC}(\varphi(x; y)) = \text{VC}\{\varphi(M; a) : a \in M^{|y|}\}.$$

2. Synthesized from multiple sources, independently worded.

Example 2.8 The formula $\varphi(x_1, \dots, x_d; w_1, \dots, w_d, b) := w_1 \cdot x_1 + \dots + w_d \cdot x_d + b \leq 0$ from the structure $(\mathbb{R}, +, \cdot, 0)$ is NIP, since half-spaces have finite VC-dimension as we will see later. 3.

Remark 2.9 ([LS13]) Saying that the structure $\mathcal{M}$ is NIP is a much stronger statement, which in general does not hold. 0.

Example 2.10 Define $\mathcal{M} := (\mathbb{R} \cup \mathcal{P}(\mathbb{R}), R)$. Let $R(a, b)$ hold for $a, b \in M$ if and only if a is in $\mathbb{R}$, b is in $\mathcal{P}(\mathbb{R})$ and $a \in b$. It is clear that $\text{VC}(x = y) = 1$, making the formula $\varphi(x; y) := x = y$ NIP. However the formula $\vartheta(x; y) := R(x, y)$ clearly has infinite VC-dimension. Hence $\mathcal{M}$ is not NIP. 2.

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As demonstrated above, disproving NIP for a structure requires finding just one formula with infinite VC-dimension. Conversely, proving that a structure or theory is NIP often requires powerful model-theoretical tools. Another key model-theoretic concept when studying localized formula behavior is the space of local types over a parameter set $A \subseteq M$.

Definition 2.11 (φ -Types [Sim15]) We use the notation φ^0 to mean $\neg\varphi$ and φ^1 to mean φ . If $\varphi(x; y)$ is a partitioned formula, a φ -type over A is a maximal consistent set of formulas of the form $\varphi(x; a)^\epsilon$, for $a \in A$ and $\epsilon \in \{0, 1\}$. The set of φ -types over A is denoted by $S_\varphi(A)$. 0.

Example 2.12 Consider the simple $\mathcal{L}$ -structure $\mathcal{M} = (\{1, 2, 3\})$, with an empty language. Let $\varphi(x; y)$ be the partitioned formula $x = y$. The formula can be constructed since equality is always part of the syntax. We will compute the set of φ -types $S_\varphi(A)$ for two different choices of parameter sets $A \subseteq M$. 3.

First, let $A = \{1\}$. Since A contains only one element, a maximal consistent set of formulas of the form $\varphi(x; a)^\epsilon$ must decide exactly whether $x = 1$ or $x \neq 1$. We obtain exactly two φ -types:

$$\begin{aligned} p_1 &= \{x = 1\}, \\ p_2 &= \{x \neq 1\}. \end{aligned}$$

However, $p_3 = \{x = 1, x \neq 1\}$ is not a φ -type, since for $x = 1$ this set of formulas is inconsistent. Thus, $S_\varphi(\{1\}) = \{p_1, p_2\}$ and $|S_\varphi(\{1\})| = 2$.

Now, let $A = \{1, 2\}$. A φ -type over A must make a consistent decision for both parameters. Specifically, it must contain exactly one of $\{x = 1, x \neq 1\}$ and exactly one of $\{x = 2, x \neq 2\}$. There are $2^2 = 4$ possible combinations, but the combination $\{x = 1, x = 2\}$ is inconsistent, as no element can be equal to both 1 and 2 simultaneously. The three consistent φ -types are:

$$\begin{aligned} q_1 &= \{x = 1, x \neq 2\}, \\ q_2 &= \{x \neq 1, x = 2\}, \\ q_3 &= \{x \neq 1, x \neq 2\}. \end{aligned}$$

Hence, $S_\varphi(\{1, 2\}) = \{q_1, q_2, q_3\}$ and $|S_\varphi(\{1, 2\})| = 3$.

To establish that a given structure is NIP—despite having infinitely many candidate formulas—one frequently restricts attention to o-minimal structures.

Definition 2.13 (O-minimality [Hie26]) Let $\mathcal{L}$ be a language containing $\mathcal{L}_<$, and let $\mathcal{M}$ be an $\mathcal{L}$ -structure such that $\mathcal{M} \models \text{DLO}$ (theory of dense linear orders). We say that $\mathcal{M}$ is *o-minimal* if every subset of M definable in $\mathcal{M}$ (with parameters) is a finite union of intervals and points. 0.

Let T be an $\mathcal{L}$ -theory such that $\text{DLO} \subseteq T$. We say T is o-minimal if every model of T is o-minimal.

Example 2.14 Let $\mathcal{M} := (\mathbb{R}, +, \cdot, \leq, 0, 1)$. By [Hie26] the theory of real closed fields (RCF) is o-minimal. Since $\mathcal{M} \models \text{RCF}$, it is o-minimal. 1.

Example 2.15 Let $\mathcal{M}' := (\mathbb{R}, +, \cdot, \sin, 0)$, where $\sin(x)$ is the known sine function. Consider the formula $\varphi(x) := \sin(x) = 0$. It is known that since $\sin$ is periodical, it has infinite zero crossings. Thus the definable set $\varphi(\mathbb{R})$ is an infinite union of points which makes the structure not o-minimal. 3.

O-minimality provides a strong structural constraint that prevents complex combinatorial behavior, directly implying NIP.

Theorem 2.16 (NIP property of o-minimal structures) *Let $\mathcal{M}$ be an $\mathcal{L}$ -structure that is o-minimal. Then it is NIP.* 3.

Proof. Assume that there is some $\mathcal{L}$ -formula $\varphi(x; y)$ such that $\mathcal{F}_\varphi$ has infinite VC-dimension. Since by definition every subset of M definable in $\mathcal{M}$ is a finite union of intervals and points, we know that each $C \in \mathcal{F}_\varphi$ is a finite union of intervals and points. Because $(M^{|x|}, \mathcal{F}_\varphi)$ has infinite VC-dimension, we know that there must be some $M' \subseteq M^{|x|}$ of infinite cardinality that is shattered. Since $\mathcal{M} \models \text{DLO}$, we know that if M' contains an interval, we can find infinite points that are not predecessors or successors of each other. If M' does not contain any interval, it has to be an infinite set of points by itself. But by definition of o-minimality, we can not shatter that infinite set of points, since every definable set is a finite union of points. $\square$

Lemma 2.17 $\mathcal{M} := (\mathbb{R}, +, \cdot, \leq, 0, 1)$ is NIP. 3.

Proof. Since $\mathcal{M}$ is o-minimal, it is NIP by Theorem 2.16. $\square$

Having seen how structural constraints like o-minimality guarantee NIP in specific models, we now return to the general framework. A foundational model-theoretic tool that will allow us to bridge local, finite conditions and global structures in the upcoming sections is the *Compactness Theorem*.

Theorem 2.18 (Compactness Theorem [Hie26]) *Let T be an $\mathcal{L}$ -theory such that every finite subset of T is satisfiable. Then T is satisfiable.* 0.

2.3 The UDTFS Conjecture and Honesty

A fundamental characterization connecting combinatorial growth with NIP formulas is given by the model-theoretic variant of the Shelah-Sauer Lemma.

Lemma 2.19 (Shelah-Sauer-Lemma [CS12]) *The following are equivalent:* 0.

1. $\varphi(x, y)$ is NIP.
2. There are $k, d \in \mathbb{N}$ such that for all finite A , $|S_\varphi(A)| \leq d \cdot |A|^k$.

Example 2.20 Example 2.12 illustrates how the number of φ -types grows 3. as we increase the size of the parameter set A , which in larger structures is bounded by the Shelah-Sauer-Lemma.

To bridge NIP theories with uniform definability properties, Chernikov and Simon isolated the *Honesty condition*. Intuitively, honesty guarantees that a formula trace on a set A can be witnessed by a global formula that does not overstep the original set outside of A .

Definition 2.21 (Honesty condition in model theory [CS12]) Let T be NIP 0. and let $\mathcal{M}$ be a model of T and $A \subseteq M$ any subset. Let $\varphi(x; a)$ have parameters in M . Then there is an elementary extension (M', A') of the pair (M, A) and a formula $\vartheta(x; b) \in \mathcal{L}(A')$ such that $\varphi(A; a) = \vartheta(A; b)$ and $\vartheta(A'; b) \subseteq \varphi(A'; a)$.

Corollary 2.22 ([CS12]) Let T, M, A and $\varphi(x; a)$ be as above. Then there 2. is $\vartheta(x; t)$ such that for any finite subset $A_0 \subseteq \varphi(A; a)$, there is $b \in A$ such that $A_0 \subseteq \vartheta(A; b) \subseteq \varphi(A; a)$.

Proof. By Definition 2.21, there exists an elementary extension $(M, A) \preceq (M', A')$ and a parameter $b^* \in A'$ such that

$$\vartheta(A; b^*) = \varphi(A; a) \quad \text{and} \quad \vartheta(A'; b^*) \subseteq \varphi(A'; a).$$

Given any finite subset $A_0 = \{a_1, \dots, a_k\} \subseteq \varphi(A; a)$, we have $A_0 \subseteq \vartheta(A; b^*)$. Moreover, since $A \subseteq A'$, the fact $\vartheta(A'; b^*) \subseteq \varphi(A'; a)$ implies that for all $x \in A$, if $x \in \vartheta(A; b^*)$, then $x \in \varphi(A; a)$.

Therefore, the structure (M', A') satisfies the first-order sentence (with parameters $a_i \in A_0$):

$$(M', A') \models \exists y \left(\bigwedge_{i=1}^k \vartheta(a_i; y) \wedge \forall x (\vartheta(x; y) \rightarrow \varphi(x; a)) \right) =: \sigma.$$

Since $(M, A) \preceq (M', A')$ is an elementary extension, this statement must already hold in (M, A) . Consequently, there exists some $b \in A$ satisfying both conditions, which yields:

$$A_0 \subseteq \vartheta(A; b) \subseteq \varphi(A; a).$$

Conversely, since we know that for any *finite* set of A_0 , any finite set of sentences σ are satisfied. This implies Definition 2.21 by Theorem 2.18, since b is obtained by the elementary extension. $\square$

Corollary 2.23 ([CS12]) *Fix a formula $\varphi(x; y)$. For every formula $\vartheta(x; t)$ (in the same variable x , but t may vary), fix an integer n_ϑ . Then there are finitely many formulas $\vartheta_1(x; t_1), \dots, \vartheta_k(x; t_k)$ such that the following holds: For every $M \models T$ and $A \subseteq M$, for every $a \in M$ there is $i \leq k$ such that for every subset $A_0 \subseteq \varphi(A; a)$ of size at most n_{ϑ_i} , there is $b \in A$ satisfying $A_0 \subseteq \vartheta_i(A; b) \subseteq \varphi(A; a)$.* 2.

Proof. Suppose for a contradiction that the statement does not hold. Then for every finite set of formulas $\{\vartheta_1, \dots, \vartheta_k\}$, there is a structure $\mathcal{M} \models T$, a subset $A \subseteq M$, and a parameter $a \in M$ such that for each $i \leq k$, there exists a finite subset $A_i \subseteq \varphi(A; a)$ of size at most n_{ϑ_i} for which no $b \in A$ satisfies $A_i \subseteq \vartheta_i(A; b) \subseteq \varphi(A; a)$.

We can express this as an $\mathcal{L}'$ -theory T' (expanding the language with a unary predicate P for A and a constant c for a), containing T together with the negation of Corollary 2.22 for every formula $\vartheta \in \mathcal{L}$.

By Corollary 2.22, no model of T' can exist, meaning T' is inconsistent. By Theorem 2.18, there already exists a finite inconsistent subset $T'_0 \subseteq T'$. This finite subset involves only finitely many formulas $\vartheta_1, \dots, \vartheta_k$. The inconsistency of T'_0 implies that for every $M \models T$, $A \subseteq M$, and $a \in M$, at least one of these ϑ_i must successfully satisfy the honest condition for any subset A_0 of size at most n_{ϑ_i} . $\square$

In [CS12], we find the following theorem, which will be omitted, since the proof is very technical:

Theorem 2.24 ([CS12]) *Let T be NIP and $\varphi(x; y)$ given. Then there is a formula $\chi(x; t)$ such that for every set A of size ≥ 2 , tuple a and finite subset $A_0 \subseteq A$, there is $b \in A$ satisfying:* 0.

1. $\varphi(A_0; a) = \chi(A_0; b)$,
2. $\chi(A; b) \subseteq \varphi(A; a)$.

We are now ready to state the UDTFs:

Definition 2.25 (UDTFS [CS12]) We say that $\varphi(x; y)$ has *Uniform Definability of Types over Finite Sets UDTFs* if there is $\vartheta(x; z)$ such that for every finite A and $a \in M$ there is $b \in A$ such that $\varphi(A; a) = \vartheta(A; b)$. We say that T satisfies UDTFs if every formula does. 0.

This leads directly to the core equivalence theorem for NIP theories.

Theorem 2.26 (UDTFS conjecture in NIP theories [CS12]) *Let T be an $\mathcal{L}$ -theory. Then it is NIP if and only if it satisfies UDTFs.* 1.

$\Rightarrow$. Let T be NIP. Then it satisfies UDTFS, by applying Theorem 2.24 and taking $A_0 = A$. $\square$

$\Leftarrow$. Let T satisfy UDTFS. By Lemma 2.19, we obtain that each formula $\varphi(x; y)$, that has UDTFS is NIP. Thus each sentence in T is NIP, proving the statement. $\square$

Lemma 2.27 *RCF is NIP and satisfies UDTFS.*

3.

Proof. We know that RCF is o-minimal, thus it is NIP by Theorem 2.16. By Theorem 2.26, we can deduce that it satisfies UDTFS. $\square$

While the UDTFS equivalence holds cleanly at the theory level, requiring an entire structure to be NIP is a very strong assumption that fails in many practical scenarios. Fortunately, in the subsequent chapters, we will observe that UDTFS can be verified locally on a per-formula basis by linking model-theoretical formulas directly to sample compression schemes in statistical learning theory.

3 Compression scheme and VC-dimension framework

3.1 VC-dimension framework

We begin by laying the foundations of statistical learning theory. First, we define a concept class, which serves as the core framework for modeling learning problems binary classification tasks.

Definition 3.1 (Concept class / Learning class [LS13]) A *concept class* or *learning class* $(X, \mathcal{C})$ consists of a pair: a domain X , and a family $\mathcal{C}$ of subsets of X . The elements of X are called points and the elements of $\mathcal{C}$ are called concepts. We will call a concept class non-degenerate if for every $x \in X$ there is a concept $C \in \mathcal{C}$ such that $x \in C$. 1.

By abuse of notation we will identify a concept with its indicator function. In other words, a concept C will also denote a function from X to $\{0, 1\}$. It is important to note that we are only considering *binary classification problems* by this construction. We call a problem binary classification problem, if the target class Y (in this case we have $Y := \{0, 1\}$) has cardinality 2. There are other types of machine learning problems, which will not be considered for simplicity. Given a sequence x of length m , we will denote by $C(x)$ the sequence: $(C(x_1), C(x_2), \dots, C(x_m)) \in \{0, 1\}^m$.

Example 3.2 (Threshold functions) 3.

$$\mathcal{C}_{tf} := (X, \mathcal{C}) = (\mathbb{R}, \{\{x \in \mathbb{R} : x \leq t\} : t \in \mathbb{R}\}).$$

Example 3.3 (Half-spaces) Fix $d \in \mathbb{N}$; 3.

$$\mathcal{C}_{hs} := (X, \mathcal{C}) := (\mathbb{R}^d, \{\{x \in \mathbb{R}^d : w \cdot x + b \leq 0\} : b \in \mathbb{R}, w \in \mathbb{R}^d\}).$$

Having introduced these classic geometric examples, it is worth clarifying a terminology distinction that frequently appears in machine learning literature: the difference between concept classes and *hypothesis classes*, since in literature, the latter term (with notation $\mathcal{H}$) is also used.

In general, a hypothesis class and a concept class need not be the same; they do not even need to be subsets of each other. Typically, hypothesis classes are understood in a more "practical" or "implementable" sense: they are the mathematical objects to which machine learning algorithms are applied, often after simplifying the theoretical concept class.

Example 3.4 For example, let $X := \mathbb{R}$ and let $\mathcal{C}$ be the set of continuous functions with exactly one zero crossing. Suppose we are only interested in the values where these functions are negative. To simplify, we might reduce the problem to considering *threshold functions*, by associating a threshold function with each function in $\mathcal{C}$. 3.

For simplicity, we will assume that $\mathcal{H} = \mathcal{C}$. To quantify the richness and expressive power of such a concept class, we introduce the central combinatorial notion of Vapnik–Chervonenkis VC-dimension via the concept of shattering.

Definition 3.5 (Shattering and VC-dimension [CS12]) The *Vapnik–Chervonenkis-dimension* (*VC-dimension*) of a concept class, $\text{VC-dim}(\mathcal{C}) := \text{VC-dim}((X, \mathcal{C}))$, is defined as the maximal cardinality of a finite set A that is shattered. If such a maximum doesn't exist, we write $\text{VC-dim}(\mathcal{C}) = \infty$.

Example 3.6 The VC-dimension of $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$ is ∞ , since $\mathbb{R}$ is clearly shattered. 3.

Example 3.7 The VC-dimension of $\mathcal{C}_{tf}$ is 1. 3.

Proof. It is clear that it is at least 1, since we can choose for some $p \in \mathbb{R}$ for example the function $x \leq p$ so that p is covered and the function $x \leq p - 1$ so that p is not covered. Hence $\{p\}$ is shattered. Assume that it is at least 2. We define the shattered set as $A := \{p_1, p_2\}$. Without loss of generality, let $p_1 < p_2$. In order for A to be shattered, there must be some threshold function, which covers p_2 but not p_1 . But any threshold function from $\mathcal{C}_{tf}$ that contains p_2 must also contain p_1 , which is a contradiction $\square$

Example 3.8 ([MRT18]) The VC-dimension of $\mathcal{C}_{hs}$ is $d + 1$. This is a fundamental result which is proven by using the *Lemma of Radon* which yields us the contradiction that we can find a point which is geometrically inside but also outside some fixed half space. 1.

In many proofs, it is useful to consider the negation or complement of a given concept. We formally define the complementary concept class and show that taking complements leaves the VC-dimension unchanged.

Definition 3.9 (Complementary concept class) Given a concept class $(X, \mathcal{C})$, we define the *complementary concept class* as: 3.

$$(X, \mathcal{C}^C) := (X, \{X \setminus C : C \in \mathcal{C}\}).$$

Example 3.10 The complementary concept class of $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$ is clearly $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$ itself. 3.

Example 3.11 The complementary concept class of $\mathcal{C}_{tf}$, whose set system are all half-open intervals of the shape $[t, \infty)$ for $t \in \mathbb{R}$, are all open intervals of the shape $(-\infty, t)$ for $t \in \mathbb{R}$ 3.

Lemma 3.12 Let $(X, \mathcal{C})$ be a concept class and $(X, \mathcal{C}^C)$ be the complementary concept class. Then $\text{VC-dim}(\mathcal{C}) = \text{VC-dim}(\mathcal{C}^C)$. 3.

Proof. We claim that a set $A \subseteq X$ is shattered by $\mathcal{C}$ if and only if it is shattered by $\mathcal{C}^C$. An immediate consequence of this claim would be clearly the statement of the lemma.

So assume that there is some set A , that is shattered by $\mathcal{C}$. Then for each arbitrary $A' \subseteq A$, there is some $C \in \mathcal{C}$ such that $A \cap C = A'$ but also there must be some $C' \in \mathcal{C}$ such that $A \cap C' = A \setminus A'$. But by Definition 3.9, we obtain that $C^C, C'^C \in \mathcal{C}^C$ with $A \cap C^C = A \setminus A'$ and $A \cap C'^C = A'$. $\square$

A crucial property of concept classes with finite VC-dimension is their stability under basic set operations. The following two classic theorems demonstrate that taking finite unions, intersections, or combining distinct classes preserves finiteness of the VC-dimension.

Theorem 3.13 ([Blu+89]) *Let $\mathcal{C} \subseteq 2^X$ be a set system of finite VC-dimension $d \geq 1$. For all $s \geq 1$, let $\mathcal{C}_s = \{\bigcup_{i=1}^s c_i : c_i \in \mathcal{C}, 1 \leq i \leq s\}$ (resp., $\mathcal{C}_s = \{\bigcap_{i=1}^s c_i : c_i \in \mathcal{C}, 1 \leq i \leq s\}$). Then for all $s \geq 1$ we obtain $VC\text{-dim}(\mathcal{C}_s) \leq 2ds \log(3s)$.* 0.

It means that the VC-dimension of the concept classes $\mathcal{C}_s$ is bounded (whose elements consist of a finite union/intersection of concepts from $\mathcal{C}$; we can use some c_i multiple times in the union or intersection which allows us to create any $\mathcal{C}_s$ with finite unions/intersections we want), hence $\mathcal{C}_s$ has also finite VC-dimension.

Another fundamental result is that the VC-dimension of the union of concept classes with the same domain which have finite VC-dimension, is finite.

Theorem 3.14 ([VW09]) *Let $\mathcal{C}_1, \dots, \mathcal{C}_m$ be set systems on a common domain X with finite VC-dimensions $V_1, \dots, V_m$, respectively. Then the VC-dimension of the set system $\bigcup_{j=1}^m \mathcal{C}_j$ is bounded.* 0.

With the basic properties of VC-dimension established, we now transition to the main focus of this chapter: compression schemes. We begin by defining samples and then introduce the formal framework for compressing data into representative sub-samples.

Definition 3.15 (Sample (set) [LS13]) *The set $S(X^m)$ of samples of size m contains all the sequences in $(X \times \{0,1\})^m$. Given a concept C and a sequence $x = (x_1, \dots, x_m)$ of size m , we will denote by $S_C(x)$ the sample S whose coordinates are $(x_i, C(x_i))$.* 0.

A naturally occurring question is whether a sample set is realizable, in other words, whether there exists a concept which is consistent with the sample set itself.

Definition 3.16 (Realizable sample set and target concept) *Let $(X, \mathcal{C})$ be a concept class. We call a sample set S of size m realizable if there exists*

2. Synthesized from multiple sources, independently worded.

a concept $C \in \mathcal{C}$ such that $C(x) = y$ for all $(x, y) \in S$. In this context, C is referred to as a *target concept* or *true concept*. If a sample set S is not realizable, we refer to this scenario as the *agnostic case*.

Remark 3.17 It is clear that for any concept $C \in \mathcal{C}$ and any sample set S of an arbitrary size $S_C(x)$ is realizable, since it is realized by C .

The fact that this must not be true is shown in the following example.

Example 3.18 Consider the concept class of threshold functions on $\mathbb{R}$. The sample set $S := \{(0, 0), (1, 1), (2, 0)\}$ is not realizable, since by definition any threshold function labeling $x \in \mathbb{R}$ as 1 must also label every $y \geq x$ as 1.

Definition 3.19 (Compression scheme [LS13]) Given a concept class $\mathcal{C}$, a k -compression scheme consists of a pair of mappings

$$\kappa: \bigcup_{m=k}^{\infty} S(X^m) \rightarrow S(X^k) \quad \text{and} \quad \rho: S(X^k) \times X \rightarrow \{0, 1\}$$

such that:

1. For any $C \in \mathcal{C}$, $m \geq k$ and $x \in X^m$, $\kappa(S_C(x))$ is a subsequence of $S_C(x)$.
2. For any $C \in \mathcal{C}$, $m \geq k$, $x \in X^m$ and a point x_i in x , $\rho(\kappa(S_C(x)), x_i) = C(x_i)$.

We call the function κ the *compression function* and the function ρ the *reconstruction function*.

We call the compression scheme *honest* if for every $C \in \mathcal{C}$, $m \geq k$, $x \in X^m$ and any $y \in X$ the following statement holds:

$$\rho(\kappa(S_C(x)), y) = 1 \implies C(y) = 1.$$

Remark 3.20 We can think of 3.19 intuitively as follows: for a fixed m we take all graphs of all possible (though not necessary existing) concepts on X^m . We choose k points from each graph we want to use as a representation for the graph itself. Since κ is not necessarily injective for $m > k$, we might end up with one subgraph representing multiple graphs on X^m . Now property 1. says that we can not "mix" graphs induced by concepts. Property 2 guarantees that for any concept on X^m we just compressed, we have to be able to tell what label our concept assigned to each point $x \in X^m$.

Example 3.21 Consider the concept class of threshold functions $\mathcal{C}_{tf}$ on $X = \mathbb{R}$. Let $S = \{(x_1, y_1), \dots, (x_m, y_m)\} \in S(\mathbb{R}^m)$ be a sample labeled by a true concept $C \in \mathcal{C}_{tf}$. We can define a 1-compression scheme (κ, ρ) as follows:

- **Compression function (κ):** Selects the single sample point with label 1 that has the maximum value:

$$\kappa(S) := \arg \max\{x_i : (x_i, y_i) \in S \text{ with } y_i = 1\}.$$

(If all points in S have label 0, κ simply selects the minimal point in S).

- **Reconstruction function (ρ):** Given the compressed sample point $(x^*, 1)$ and a query point $x \in \mathbb{R}$, ρ sets the decision boundary precisely at x^* :

$$\rho((x^*, 1), x) := \begin{cases} 1 & \text{if } x \leq x^*, \\ 0 & \text{if } x > x^*. \end{cases}$$

Remark 3.22 Sample compression schemes are implicitly defined over realizable sets, as they rely on the existence of a true concept C for the sets $S_C(x)$. Hence, no guarantees are provided for the agnostic case. 3.

Definition 3.23 (Bounded compression scheme) We call a compression scheme *bounded*, if there is a $C \in \mathbb{N}$, such that for any m and $x \in X^m$ there is a k -compression scheme as in Definition 3.19 with $k \leq c$. 3.

Remark 3.24 Note that definitions 3.19 and 3.23 are equivalent, since a k -compression scheme is bounded by definition (by k itself) and a bounded compression scheme can be extended to c -compression scheme, by simply choosing (if needed) more points we want a set to be compressed on. 3.

To prove the main result of Livni and Simon, standard compression schemes are often too rigid. We therefore consider natural generalizations that allow for additional side information or fixed symbols, such as an extra bit.

Definition 3.25 (Compression scheme with additional information [LS13]) Given a finite set Q , a *k-compression scheme with additional information* Q consists of a pair of functions ρ and κ , as in the previous definition, with the modification that the image of κ is $Q \times S(X^k)$ and the domain of ρ is $Q \times S(X^k) \times X$. Property 1 is replaced by the following requirement: let $\kappa(S_C(x)) := (q, x_\kappa)$. x_κ has to be a subsequence of $S_C(x)$. 0.

It is often convenient to encode the additional information via reordering of the subsequence or using a bounded amount of repetitions of certain elements. In our case we will use the following type of a compression scheme, which can be considered a special case of a compression scheme with additional information.

Definition 3.26 (Word compression scheme with an extra bit b [LS13]) A *k-word compression scheme with an extra bit b* consists of a pair of functions 0.

ρ and κ , as in the first definition, with the modification that the image of κ is $(b \cup X)^k$ and the domain of ρ is $(b \cup X)^k \times X$.

We replace property 1 with the requirement that all elements in the sequence $\kappa(S_C(x))$ are either the extra bit b or elements in the sequence x .

Theorem 3.27 (Definition hierarchy [LS13]) *Definition 3.26 is a special case of Definition 3.25, and thus a k -compression scheme with an extra bit b induces a k -compression scheme with additional information.* 2.

Proof. It is sufficient to prove that some k -compression scheme with extra bit induces some bounded compression scheme with additional information due to our discussion after Definition 3.23.

Assume we have a k -compression scheme with an extra bit b as in Definition 3.26, which yields us a compression function κ and a reconstruction function ρ .

First, we define $Q := \{1, \dots, k, b\}^k$, which we can think of as a set which contains all possible "blueprints" how κ maps $S_C(x)$ onto a word which contains elements of x and the bit b . Since κ compresses on a word of length k by definition, we have a finite amount of combinations how $S_C(x)$ can be compressed: we have k spots for a word, which can be filled with datapoint-label tuples or the additional bit b . Thus we can bridge Q and the properties of κ as follows: consider $q \in Q$. If q_i is a number, that means that κ maps $S_C(x)$ on a word where on position i there is an element of x . Else on position i of the word $\kappa(S_C(x))$ is the extra bit. Q contains $(k+1)^k$ elements, which means that it is a finite set.

We construct a compression function κ' as in Definition 3.25. Choose $x \in X^m$ for some $k \leq m$ and $C \in \mathcal{C}$, consider $w := \kappa(S_C(x))$. Define x_κ as the subsequence of w , which contains only elements from x and take $q_x \in Q$, which matches the word $\kappa(S_C(x))$ as a blueprint like discussed above. It is clear that modified condition 1 is satisfied by our construction.

Note that we constructed a bijection h from the words w to the tuples (q_x, x_{kappa}) . For injectivity, take some words w and w' with $(q_x, x_{kappa}) = h(w) = h(w') = (q'_x, x'_{kappa})$ and $w \neq w'$. We show that this assumption yields us a contradiction. We have to distinguish two cases:

1. We find some index i where $w_i = b$ and $w'_i = (x, y)$ where (x, y) is some datapoint-label tuple. But then we obtain immediately that $q_x = b$ and $q'_x = n$ where n is some natural number with $n \leq k$, hence $h(w) \neq h(w')$.
2. We find some index i where $w_i = (x, y)$ and $w'_i = (x', y')$ where (x, y) and (x', y') are some different datapoint-label tuples. Now exactly two things can happen. Without loss of generality, $(x, y) \notin w'$, then we would obtain that $x_{kappa} \neq x'_{kappa}$. Or we would find some index $j \neq i$, such that $w_j = (x', y')$ and $w'_j = (x, y)$. But since x_{kappa} and x'_{kappa} are

subsequences, we would find that both subsequences have a different order of (x, y) and (x', y') . So we obtain again that $x_{kappa} \neq x'_{kappa}$. In both cases this leads to $h(w) \neq h(w')$.

The surjectivity holds by construction of sample compressions, since we obtain every possible combination of words by definition of κ in 2, and hence every possible tuple (q_x, x_{kappa}) . Thus, there is some inverse map h^{-1} .

The reconstruction function is defined as

$$\rho'(\kappa'(S_C(x)), x_i) := \rho(h^{-1}(\kappa'(S_C(x))), x_i) = \rho(\kappa(S_C(x)), x_i).$$

It is clear that ρ' satisfies 2, since ρ and κ came from a k -compression scheme with an extra bit b. $\square$

3.2 Practical Example: Spam Classification

To illustrate our learning-theoretic framework in a practical setting, consider the task of classifying emails as *spam* or *not spam*. We model this problem by focusing on a small set of indicative keywords, such as "win" and "price". While real-world spam detection systems rely on far more sophisticated features, this simplified approach suffices to demonstrate the core ideas.

Each email is represented as a vector in $\mathbb{R}^2$, where the coordinates correspond to the counts of the respective keywords it contains. Clearly, not every email containing these keywords is spam, so we define a decision boundary using a hyperplane in $\mathbb{R}^2$. Emails whose vectors lie above this hyperplane are classified as spam, and those below as non-spam. Thus, our concept class is the class of half spaces in $\mathbb{R}^2$, which is known to be PAC-learnable, since it has finite VC-dimension. Consequently, we can compute a good approximation for classifying emails based on this simple approach.

Our goal is to minimize the *linear-least-squares-error* (LLS), which measures the discrepancy between the predicted and actual classifications, based on some sample set of size m . Formally, given a sample of m emails, where each email i is represented by a feature vector $(a_{i1}, a_{i2}) \in \mathbb{R}^2$ and labeled with $y_i \in \{0, 1\}$ (where 1 indicates spam and 0 indicates non-spam), the LLS is defined as follows:

Definition 3.28 (Linear Least Squares Error) The linear least squares error (LLS) is defined as:

$$\text{LLS} := \sum_{i=1}^m \left(\sum_{j=1}^2 a_{ij} w_j - y_i \right)^2,$$

where $w = (w_1, w_2) \in \mathbb{R}^2$ represents the weights of the hyperplane, and the a_{ij} are the feature values of the i -th email.

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In this setting, we assume that our sample set of m emails is drawn independently and identically distributed from a larger pool of collected emails and the sample set is realizable. To find the optimal weights w that minimize the LLS, we can use the known solution for this problem:

$$w = (X^T X)^{-1} X^T y,$$

where $X \in \mathbb{R}^{m \times 3}$ is the data matrix with the first two columns corresponding to the features of the datapoints, the last one being a column filled with ones to calculate the bias and $y \in \mathbb{R}^m$ is the vector of labels. It is important to know, that the first coordinate obtained by solving this problem will be the earlier discussed bias of our hyperplane and the last ones will be the weight coordinate w_1, w_2 .

However, if we knew the exact honest k -sample compression scheme for this concept class, we could avoid solving this (in higher dimensions costly due to inverting $X^T X$) optimization problem entirely. Instead, given a sample of size at least k , we could directly derive a concept (in this case a hyperplane in $\mathbb{R}^2$) from the compression scheme. This would yield a half space that correctly classifies the training set. Moreover, due to the honesty property, we would be guaranteed that if the model classifies an email as spam, it would indeed be spam in reality, assuming the training data is representative of the true distribution.

3.3 Homogeneous and Alternating Sequences

To control the behavior of long sequences in arbitrary concept classes, we rely on structural regularity results from Ramsey theory. Specifically, Ramsey's Theorem for Hypergraphs guarantees that sufficiently long sequences contain highly symmetric sub-structures.

Theorem 3.29 (Ramsey's Theorem for Hypergraphs [GRS90]) *For all positive integers l , c , and q , there exists an integer n_0 such that for all $n \geq n_0$, if the set of q -element subsets $[n]^q$ is c -colored, there exists a subset of size l whose q -element subsets $[l]^q$ are entirely monochromatic.*

Remark 3.30 (Notations)

- By $[n]$ with $n \in \mathbb{N}$ we mean the set $\{1, \dots, n\}$
- By $[n]^q$ with $q \in \mathbb{N}$ we mean $\{X \subseteq [n] : |X| = q\}$

In other words, if we have a set in which for each color there is at least one element painted in that color, we can find a sufficiently large integer such that there must be a subset of size l such that all q -element subsets are colored in a single color. However, we do not know which color that will be.

Using Ramsey's Theorem as our main tool, we now introduce the concept of d -homogeneous sequences. Homogeneity forces all subsequences of a fixed length d to exhibit identical labeling patterns under the concept class.

Definition 3.31 (Homogeneous sequence [LS13]) Let $(X, \mathcal{C})$ be a concept class. A sequence x of elements in X is called *d-homogeneous* if for any two subsequences of size d , denoted $x^{(1)}$ and $x^{(2)}$, and for any concept $C_1 \in \mathcal{C}$, there is a concept $C_2 \in \mathcal{C}$ such that

$$C_1(x^{(1)}) = C_2(x^{(2)}).$$

Example 3.32 To visualize *d-homogeneity*, consider points $x_1 \leq x_2 \leq x_3 \leq x_4$ on $\mathbb{R}$ with threshold functions $\mathcal{C}_{tf}$. If we look at 2-subsequences, the sequence is homogeneous because any labeling pattern achievable on (x_1, x_2) can be mirrored on (x_3, x_4) by shifting the threshold.

Lemma 3.33 Let $(X, \mathcal{C})$ be a concept class and x a *d-homogeneous sequence* of elements in X . Then each subsequence x' of x of length of at least d is also *d-homogeneous*.

Proof. Assume that there is a subsequence $\tilde{x}$ of x of length of at least d that is not *d-homogeneous*. That means that there are two subsequences of size d , $\tilde{x}_1$ and $\tilde{x}_2$ such that there is a concept $\tilde{C}_1 \in \mathcal{C}$ such that for every $\tilde{C}_2 \in \mathcal{C}$ we get

$$\tilde{C}_1(\tilde{x}_1) \neq \tilde{C}_2(\tilde{x}_2). \quad (3.1)$$

But by definition $\tilde{x}_1$ and $\tilde{x}_2$ are subsequences of x , each of size d . Since x is *d-homogeneous* by assumption, (3.1) can not hold. $\square$

While *d-homogeneity* is a strong condition, the next fundamental theorem shows that every sufficiently long sequence is guaranteed to contain a large *d-homogeneous* subsequence.

Theorem 3.34 ([LS13]) For any k and d , there exists a large enough p such that: Given a concept class $(X, \mathcal{C})$, any sequence x of size p has a *d-homogeneous* subsequence of size k .

Proof. We apply Ramsey's Theorem 3.29 in the following way. Define $c := 2^{2^d}$, because we want to get each constellation of each combination of a labeled set $[d]$. So we create $\mathcal{P}([d])$, first, to get every possible combination of a 0, 1-label vector. Now, we need to get every constellation of a label vector; so let us say $d_{constellation}$ is a constellation then each $d_{combination} \in d_{constellation}$ is some 0, 1-label vector. We get the constellation by observing $\mathcal{P}(\mathcal{P}([d]))$. This cardinality is known as 2^{2^d} . So each color represents a constellation we are able to achieve through the given concept class $(X, \mathcal{C})$. We recall that the underlying set we apply Ramsey's Theorem are datapoints from X . So if we paint some set with d elements in a certain color, we know that for each $d_{combination} \in d_{constellation}$, we find some $C_{combination} \in \mathcal{C}$, which yields this label vector in the exact order $d_{combination}$.

Since we are looking for a *d-homogeneous* sequence, we have to consider sets of cardinality d , hence we define $q := d$. We say $l := k$ in consequence, because we are looking for a sequence of size k .

Ramsey's Theorem yields us $p := n_0$, after we used each color at least once. We now find a set of size k whose d -element subsets $[k]^d$ are entirely monochromatic. So we know that if we choose an arbitrary d -element subset D_1 with some fixed combination $D_{\text{combination}} \in D_1$ and we choose another arbitrary d -element subset D_2 , we will find $D_{\text{combination}} \in D_1$. Clearly those sets correspond with (sub-)sequences and the combinations with some concepts. Since this argumentation holds for any sequence we can obtain from $[k]^d$, we have proven this theorem for any sequence x of size p . $\square$

Definition 3.35 (Alternation number [LS13]) Given a concept class $(X, \mathcal{C})$, a sequence y is said to be *alternating* in $\mathcal{C}$ if there is a concept $C \in \mathcal{C}$ such that $C(y_i) \neq C(y_{i-1})$ for every $1 < i \leq |y|$. 0.

The *alternation number* of a sequence x in $\mathcal{C}$, denoted $\text{alt}(x; \mathcal{C})$, is the size of the largest subsequence of x that is alternating in $\mathcal{C}$. Formally,

$$\text{alt}(x; \mathcal{C}) := \max_{C \in \mathcal{C}} |\{i : C(x_i) \neq C(x_{i-1})\}| + 1.$$

Theorem 3.36 ([LS13]) Let $(X, \mathcal{C})$ be a concept class of VC-dimension d and let x be a $(d+1)$ -homogeneous sequence, then 1.

$$\text{alt}(x; \mathcal{C}) \leq 2d + 1.$$

Proof. We will prove the theorem, by assuming that x is a $(d+1)$ homogeneous sequence with

$$\text{alt}(x; \mathcal{C}) \geq 2d + 2,$$

which yields the contradiction that

$$\text{VC-dim}(\mathcal{C}) \geq d + 1.$$

Without loss of generality, we may assume that x is a sequence of size $2d+2$, and there is a concept $C \in \mathcal{C}$ such that $C(x_i) = 1$ if and only if i is even.

We can assume the former, since $\text{alt}(x; \mathcal{C}) \geq 2d+2$ means that there is $C' \in \mathcal{C}$, such that C' causes at least $2d+1$ label changes on x . Hence it is easy to see, that x must be of size of at least $2d+2$. If x is of bigger size, we simply use Lemma 3.33, so our argumentation for those cases is still valid.

Since it is sufficient to assume that x is of size $2d+2$, and $\text{alt}(x; \mathcal{C}) \geq 2d+2$, it is also easy to see that there has to be $C \in \mathcal{C}$ which labels even and odd elements respectively onto two different classes. Hence we can assume the latter (if there is only a C which yields $C(x_i) = 1$ if and only if i is odd, we get the same argumentation but for 0).

Given a boolean sequence $b \in \{0, 1\}^{d+1}$, we construct a subsequence y of x such that $C(y) = b$.

Choose $y_1 = x_1$ if $b_1 = 0$ and $y_1 = x_2$ otherwise. We define y_n inductively for every $n \leq d+1$. Assume we chose x_i to be y_{n-1} . Let $y_n = x_{i+2}$ if $b_n = b_{n-1}$, and $y_n = x_{i+1}$ if $b_n \neq b_{n-1}$. Intuitively speaking, we can tell at

this point, that at most every two elements, we will find the one we need from x to construct our sequence y .

By homogeneity, there exists a concept C_2 such that $C_2(x_1, \dots, x_{d+1}) = C(y) = b$. As b was arbitrary, $\{x_1, \dots, x_{d+1}\}$ is shattered. $\square$

Combining the subsequence extraction result (Theorem 3.34) with the bounded alternation of homogeneous sequences (Theorem 3.36) yields the following crucial corollary, which will be essential for constructing our compression schemes.

Corollary 3.37 ([LS13]) *For any k and d , there exists a large enough p such that, given a concept class $(X, \mathcal{C})$ with $VC\text{-dim}(\mathcal{C}) = d$, any sequence x of length p has a subsequence y of length k such that*

$$\text{alt}(y; \mathcal{C}) \leq 2d + 1.$$

Proof. Choose d as $d + 1$ in 3.34 and apply 3.36 on a $d + 1$ homogenous subsequence y . $\square$

3.4 Alternating Concept Classes and Honest Concepts

In order to formalize the main result of this chapter, we need to define and understand the machinery of *alternating concept classes*.

Definition 3.38 (Alternating concept class [LS13]) Given a concept class $(X, \mathcal{C})$ and a sequence x such that $|x| = m$, we consider the following subset of X for every $i \leq m$:

$$\text{ALT}(x; \mathcal{C}) := \{y \in X : \text{alt}((y, x_1, \dots, x_m); \mathcal{C}) \leq 2 \text{VC-dim}(\mathcal{C}) + 1\}.$$

Given m and a subset $A \subseteq X$ (possibly infinite), we define the family of subsets:

$$\mathcal{ALT}_m(A; \mathcal{C}) := \{\text{ALT}(x; \mathcal{C}) : x \in A^m\}.$$

The concept class $(X, \mathcal{ALT}_m(A; \mathcal{C}))$ is called the *alternating concept class of $(X, \mathcal{C})$ with respect to sequences of length m from A* .

Remark 3.39 We want to have a better intuition for $\mathcal{ALT}_m(A; \mathcal{C})$, since it is a difficult but very important one. For that, let us fix an $m \in \mathbb{N}$. Depending on our m , $\mathcal{ALT}_m(A; \mathcal{C})$ looks different;

1. We choose $m \leq 2\text{VC-dim}(\mathcal{C})$. We get $\mathcal{ALT}_m(A; \mathcal{C}) = \{X, \dots\}$, because we can add each $y \in X$ in front of any sequence x of length m . The reason for that is that each concept c of our concept class can cause at most $m + 1 \leq 2\text{VC-dim}(\mathcal{C}) + 1$ label changes on our sequence (y, x) . Hence any sequence of size m will create the set X .

2. We choose $m > 2\text{VC-dim}(\mathcal{C})$. Now the situation will get a bit more complicated depending on the sequence. Exactly three situations can occur:

- (a) There is a concept c_1 , which causes more than $2\text{VC-dim}(\mathcal{C}) + 1$ label changes on x . This can happen if we choose for instance $m > 2\text{VC-dim}(\mathcal{C}) + 4$ and we have two datapoints $y_1, y_2 \in A$ such that c_1 labels those two points differently. If now choose

$$x := \underbrace{y_1, y_2, \dots, y_1, y_2, y_1}_{\text{length } m}$$

This is possible, since we are allowed to use points multiple times and in any order for our sequence x . Now we know that we can not put any point y in front of our sequence x , because c_1 already causes more than $2\text{VC-dim}(\mathcal{C}) + 1$ label changes on x itself. Hence this x creates the empty set. So we get a concept, which labels all points from X onto 0.

- (b) The concept $c_2 \in \mathcal{C}$, which causes the most changes, causes exactly $2\text{VC-dim}(\mathcal{C}) + 1$ label changes on x . Then x induces c_2 if and only if $c_2(x_0) = 1$ and equivalently x induces $X \setminus c_2$ if and only if $c_2(x_0) = 0$.
- (c) The concept $c_3 \in \mathcal{C}$, which causes the most changes, causes less than $2\text{VC-dim}(\mathcal{C}) + 1$ label changes on x . Then x induces X , with the analogue argumentation as in case 1.

1. The set that is induced by x is not unique, since there can be multiple concepts which cause the most label changes on the sequence.
2. Even though we took $A \subseteq X$, we get that $\mathcal{ALT}_m(A; \mathcal{C}) \subseteq \mathcal{P}(X)$.
3. The alternating concept class does not need to be a subset of our original concept class! The converse in general also does not hold.

Why are alternating concept classes interesting, though especially regarding Remark 3? The alternating concept class $(X, \mathcal{ALT}_{2d+1}(X; \mathcal{C}))$ will play a crucial role in formalizing our main results.

Analyzing the cases in Remark 3.39 yields a key structural property: for sequences of length $2d+1$, the alternating concept class simplifies drastically into a union of base concept classes.

Corollary 3.40 *Let $(X, \mathcal{C})$ be a concept class with VC-dimension d . Then $\mathcal{ALT}_{2d+1}(X; \mathcal{C}) \subseteq \mathcal{C} \cup \mathcal{C}^C \cup X$, where $\mathcal{C}^C$ are the concepts from the complementary concept class (see Definition 3.9).* 3.

Proof. Let $x \in X^{2d+1}$ be an arbitrary sequence and $y \in X$ an arbitrary point. Then clearly any $C \in \mathcal{C}$ can cause at most $2d+1$ label changes on the sequence (y, x) , thus either case 2b or case 2c happen. So x induces either some $C \in \mathcal{C}$, $C \in \mathcal{C}^C$ or X , proving the corollary. $\square$

By combining the structural decomposition from 3.40 with the stability properties of finite unions and complements, we can deduce that the VC-dimension of the alternating concept class remains strictly bounded.

Theorem 3.41 (Boundary of alternating concept class VC-dimension) *Let $(X, \mathcal{C})$ be a concept class with VC-dimension d and $(X, \mathcal{ALT}_{2d+1}(X; \mathcal{C}))$ its alternating concept class. Then $(X, \mathcal{ALT}_{2d+1}(X; \mathcal{C}))$ has finite VC-dimension.* 3.

Proof. Because of Corollary 3.40 and Theorem 3.14, we know that $\text{VC-dim}(\mathcal{ALT}_{2d+1}(X; \mathcal{C}))$ is bounded by $\text{VC-dim}(\mathcal{C})$, $\text{VC-dim}(\mathcal{C}^C)$ and $\text{VC-dim}(\{X\})$. Since $\text{VC-dim}(\mathcal{C})$ is bounded by assumption, we know that $\text{VC-dim}(\mathcal{C}^C)$ is also bounded by Lemma 3.12. Clearly $\text{VC-dim}(\{X\})$ is also bounded by 1. Thus $(X, \mathcal{ALT}_{2d+1}(X; \mathcal{C}))$ has finite VC-dimension. $\square$

Lemma 3.42 ([LS13]) *If $\mathcal{ALT}_{2\text{VC-dim}(\mathcal{C})+1}(X; \mathcal{C})$ has finite VC-dimension, then $\mathcal{ALT}_m(X; \mathcal{C})$ has finite VC-dimension for every m .* 1.

Proof. Let $(X, \mathcal{C})$ be a concept class with $\text{VC-dim}(\mathcal{C}) = d$. Given $m \in \mathbb{N}$ and $y \in X$. If $y \notin \text{ALT}(x; \mathcal{C}) \neq \emptyset$ for some $\text{ALT}(x; \mathcal{C}) \in \mathcal{ALT}_m(X; \mathcal{C})$, then x has a subsequence $\tilde{x} = (x_{i_1}, x_{i_2}, \dots, x_{i_{2d+1}})$ such that

$$\text{alt}((y, x_{i_1}, x_{i_2}, \dots, x_{i_{2d+1}}); \mathcal{C}) = 2d + 2.$$

Note that the converse is trivially true, since if there is a concept which causes too many label changes on $(y, \tilde{x})$, we know that this concept will of course cause at least as many label changes on (y, x) as it does on the subsequence.

In other words, $y \notin \text{ALT}(x; \mathcal{C})$ if and only if x has a subsequence $\tilde{x}$ such that

$$y \notin \text{ALT}((x_{i_1}, x_{i_2}, \dots, x_{i_{2d+1}}); \mathcal{C}).$$

which is equivalent to $y \in \text{ALT}(x; \mathcal{C})$ if and only if for all subsequences $\tilde{x}$ of x we get that

$$y \in \text{ALT}((x_{i_1}, x_{i_2}, \dots, x_{i_{2d+1}}); \mathcal{C}).$$

Since we obtained

$$\text{ALT}(x; \mathcal{C}) = \bigcap_{\substack{\tilde{x} \text{ is a subsequence of } x \\ |\tilde{x}|=2d+1}} \text{ALT}(\tilde{x}; \mathcal{C})$$

and thus $\text{ALT}(x; \mathcal{C}) \in \mathcal{ALT}_m(X; \mathcal{C})$ is a finite intersection of elements $\text{ALT}(\tilde{x}; \mathcal{C}) \in \mathcal{ALT}_{2\text{VC-dim}(\mathcal{C})+1}(X; \mathcal{C})$. By the theorem for new concept classes created by

finite intersections of elements from the old concept class; see 3.13, we get that $\mathcal{ALT}_m(X; \mathcal{C})$ has finite VC-dimension for every m , since m was chosen arbitrarily. $\square$

Having established that alternating concept classes inherit finite VC-dimension for arbitrary sequence lengths m , we now introduce the combinatorial notion of honesty. This condition guarantees that compressed or reconstructed hypotheses do not make false positive errors on the training set.

Definition 3.43 (Honesty condition [LS13]) Given a concept $C \in \mathcal{C}$ and a finite set $A \subseteq X$, we say that a set $\tilde{C} \subseteq X$ (not necessarily a concept in $\mathcal{C}$) is *honest* for A with respect to C if:

$$\tilde{C} \cap A \subseteq C \cap A.$$

Definition 2.21 is the model-theoretic analogue (and a substantial generalization) of Definition 3.43. To see this connection, we have to define R -structures.

Definition 3.44 (R -Structures and formulas [LS13]) An R -structure is a (usually infinite) set M equipped with a distinguished binary relation $R(x, y)$. An R -formula is a syntactically correct formula of the R -structure.

Remark 3.45 ([LS13]) How does this framework relate to concept classes? We have explained how, given a structure M , one associates various concept classes $\mathcal{F}_\varphi$ on cartesian powers of M . Conversely, starting with a concept class $(X, \mathcal{C})$, we can construct an R -structure M as follows. Let the set M consist of the disjoint union of the sets X and $\mathcal{C}$. Let $R(a, b)$ hold for $a, b \in M$ if and only if a is in X , b is in $\mathcal{C}$ and a belongs to b .

Now consider a concept class $(X, \mathcal{C})$ embedded into an R -structure as discussed earlier. Then we obtain the property $\vartheta(A; b) \subseteq \varphi(A; a)$, where b corresponds with $\tilde{C}$, a corresponds with C , the formulas ϑ and φ encode the intersections by using the relation R and the resulting definable sets encode the elements that are in both of the intersected sets.

Remark 3.46 While the combinatorial Definition 3.43 merely requires the inclusion $\tilde{C} \cap A \subseteq C \cap A$ for a single set A , the model-theoretic condition (Definition 2.21) is significantly stronger:

1. **Exact trace matching on A :** It requires the trace on A to be *identical*; $\vartheta(A; b) = \varphi(A; a)$ (which means $\vartheta(A; b) \subseteq \varphi(A; a)$ and $\varphi(A; a) \subseteq \vartheta(A; b)$ simultaneously).
2. **Uniformity over extensions:** The inclusion condition $\vartheta(A'; b) \subseteq \varphi(A'; a)$ does not merely hold on A , but is preserved in an elementary extension $(\mathcal{M}'; A')$. This guarantees that $\vartheta(x; b)$ is globally constrained by $\varphi(x; a)$ across all elements that "behave like A " in the

model-theoretic sense, preventing ϑ from picking up spurious elements outside A .

Thus, Definition 2.21 ensures that the local behavior on A is witnessed by a formula ϑ that is globally honest with respect to φ .

Definition 3.47 (Honest alternating concept class [LS13]) Let $(X, \mathcal{C})$ be a concept class, $A \subseteq X$ a finite set, and $C \in \mathcal{C}$ a concept. The concept class $(X, \mathcal{ACT}_m(A, C; \mathcal{C}))$ consists of all concepts in $\mathcal{ACT}_m(A; \mathcal{C})$ that are honest for A with respect to C . 0.

3.5 The (p, k) Theorem

To transition from sequence-based label changes to finite coverage guarantees, we require a combinatorial tool capable of bounding covering numbers globally. We introduce the (p, k) property, which formalizes local intersection densities.

Definition 3.48 ((p, k) property [LS13]) Given a concept class $(X, \mathcal{C})$, a finite set $A \subseteq X$ is said to have the (p, k) property with respect to $(X, \mathcal{C})$ if for every subset $A_0 \subseteq A$ of cardinality p , the intersection $A_0 \cap C$ has a cardinality of at least k for some concept $C \in \mathcal{C}$. 0.

Remark 3.49 While the honesty property for finite subsets 2.22 and the combinatorial (p, k) -property 3.48 (as used by Livni et al.) are formally distinct notions, they play functionally analogous roles in their respective frameworks: 3.

- In learning theory, the (p, k) -property serves as the key combinatorial bridge to construct *honest sample compression schemes* from classes of bounded VC-dimension.
- In model theory, the honesty property for finite subsets acts as the model-theoretic counterpart: it uses compactness and elementarity to reduce the global honesty condition to finite subsets A_0 , forming the core building block for establishing UDTFS and definable compression schemes in NIP theories.

Thus, both concepts serve as the crucial structural pivot to transition from bounded complexity (NIP / VC-dimension) to local, parameter-efficient reconstructions (UDTFS / Compression). As we will see in the following, we will establish similarly the $(p, k_1, \dots, k_d)$ -Theorem for bounded VC-dimension concept classes (see 3.54) regarding the model-theoretical context.

Equipped with this intuition, we now state the main (p, k) -Theorem for concept classes of bounded VC-dimension, which guarantees the existence of a uniform finite cover.

Theorem 3.50 ((p, k) -Theorem for bounded VC-dimension concept class [LS13]) *For every k and p , there exists a constant N such that the following holds: Given a non-degenerate concept class $(X, \mathcal{C})$ with $VC\text{-dim}(\mathcal{C}) < k$, if a finite subset $A \subseteq X$ has the (p, k) property with respect to $(X, \mathcal{C})$, then there exist N concepts $C_1, \dots, C_N \in \mathcal{C}$ such that:* 1.

$$A \subseteq \bigcup_{i=1}^N C_i.$$

At first glance, one might attempt to establish this uniform bound N by applying Ramsey's Theorem (3.29) again, setting $l := k$, $q := p$, and $c := 2$. We could color a set A_0 with p elements as 1, if there is a $C \in \mathcal{C}$ such that $k \leq |C \cap A_0|$ and with color 2 otherwise. This would yield us some n and we would find a monochromatic set. If this set were monochromatic with color 2, we could discard it, since it means that it does not fulfill the (p, k) -property and re-apply Ramsey. If we find a monochromatic set M colored with color 1, then we would get our $N := \binom{|M|}{k}$, since we could just take those C we found when we colored the set A_0 with color 2. However, the key obstacle is that the resulting N depends directly on the cardinality of the monochromatic set M . But the (p, k) -Theorem needs some bound N for any set that satisfies the (p, k) property. Suppose that for any $M_0 \subset X$ with $|M_0| = c_0$, we find some $M_1 \subset X$ with $|M_1| = c_1$ and $c_0 < c_1$. This means that our N we will find while re-applying Ramsey will be unbounded. This does not contradict the fact that the VC-dimension of our concept class is smaller than k .

Thus Ramsey-theoretic arguments only provide local structure that scales with the domain size, while we require a global, size-independent bound N . So we need another approach.

The (p, k) -Theorem originates from the following result by Matoušek. To maintain a clear focus on the learning- and model-theoretic core of this thesis, the details of the reduction will not be further pursued.

Theorem 3.51 (Matoušek's (p, k) -Theorem [Mat04]) *Let $\mathcal{F}$ be a set system with $\pi_{\mathcal{F}}^*(m) = o(m^k)$ for some integer k , and let $p \geq k$. Then there is a constant T such that the following holds for every finite family $\mathcal{G} \subseteq \mathcal{F}$: If $\mathcal{G}$ has the (p, k) -property, meaning that among every p sets of $\mathcal{G}$, some k intersect, then $\tau(\mathcal{G}) \leq T$.* 0.

3.6 Compression schemes and finite VC-dimension

We now arrive at the central synthesis of this chapter: proving that concept classes with finite VC-dimension admit bounded compression schemes. The following main theorem serves as the core bridge between combinatorial learnability and model-theoretic stability.

Theorem 3.52 ([LS13]) *Let $(X, \mathcal{C})$ be a concept class with $VC\text{-dim}(\mathcal{C}) = d$ and assume that $(X, \mathcal{AL}\mathcal{T}_{2d+1}(X; \mathcal{C}))$ has finite VC-dimension. Then $(X, \mathcal{C})$ has a bounded compression scheme with additional information.*

To establish Theorem 3.52, the standard (p, k) -property is insufficient because alternating concept classes conditioned on honest traces may fail to satisfy it globally. We therefore introduce a multi-class generalization, the $(p, k_1, k_2, \dots, k_d)$ property.

Definition 3.53 ($(p, k_1, k_2, \dots, k_d)$ property [LS13]) Let $(X, \mathcal{C}) = (X, \mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_d)$ be d concept classes sharing a common domain X . A subset $A_0 \subseteq X$ of size p is said to *suggest property (p, k_n)* with respect to $\mathcal{C}_n$ if $A_0 \cap C_n$ is of cardinality at least k_n for some concept $C_n \in \mathcal{C}_n$.

A subset $A \subseteq X$ is said to have the $(p, k_1, k_2, \dots, k_d)$ property with respect to $(X, \mathcal{C})$ if every subset $A_0 \subseteq A$ of size p suggests property (p, k_n) with respect to $\mathcal{C}_n$ for some $n \leq d$.

Originally in [LS13], where this definition originates from, the (p, k) -property was defined with A being finite while here it was defined to not necessarily be finite. Both properties can be defined with infinite sets A respectively, however we will keep everything as it was in the original paper.

It is a relaxation, because if A has the (p, k) property, then we define $k_1 := k$ and choose arbitrary $k_2, \dots, k_d \in \mathbb{N}$. By definition now every subset $A_0 \subseteq A$ of size p suggests property (p, k_1) with respect to $\mathcal{C}_1$, and hence A satisfies the $(p, k_1, k_2, \dots, k_d)$ property.

We now extend Matoušek's covering result to this multi-class setup, demonstrating that the relaxed $(p, k_1, \dots, k_d)$ property still yields a global finite cover.

Theorem 3.54 ($(p, k_1, \dots, k_d)$ Theorem for bounded VC-dimension concept classes [LS13]) *For every $p, k_1, \dots, k_d$, there exists a constant N such that the following holds: Given d non-degenerate concept classes sharing a common domain $(X, \mathcal{C}) = (X, \mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_d)$ with $VC\text{-dim}(\mathcal{C}_i) < k_i$ for every $i \leq d$, if a finite subset $A \subseteq X$ has the $(p, k_1, \dots, k_d)$ property with respect to $(X, \mathcal{C})$, then there exist N concepts $\{C_1, \dots, C_N\} \subseteq \bigcup_{i=1}^d \mathcal{C}_i$ such that:*

$$A \subseteq \bigcup_{j=1}^N C_j.$$

Proof. The concept class $(X, \bigcup_{i=1}^d \mathcal{C}_i)$ is the union of d concept classes with finite VC-dimension, hence its VC-dimension is bounded by Theorem 3.14, say by D (a bound D that is dependent on $k_1, \dots, k_d$ may be produced). We prove that the $(p, k_1, \dots, k_d)$ -property implies the $(P, D + 1)$ -property with respect to $(X, \bigcup_{i=1}^d \mathcal{C}_i)$ for a P chosen sufficiently large. Then we can deduce from Theorem 3.50, that there are N concepts $c_1, \dots, c_N \in$

$\bigcup_{i=1}^d \mathcal{C}_i$, such that $A \subseteq \bigcup_{i=1}^N \mathcal{C}_i$. Assume that an arbitrary set A satisfies the $(p, k_1, \dots, k_d)$ -property and not the $(P, D+1)$ -property. Define $N := \max\{N_1, \dots, N_d\}$, where the bounds $N_1, \dots, N_d$ are the ones we obtain by applying Theorem 3.50 on each $A_i \subseteq A$ which satisfy the (p, k_i) -property for some i with respect to a concept class with $\text{VC-dim}(\mathcal{C}_i) < k_i$ and hence yield us those bounds.

We now create a multigraph out of A by coloring $A_0 \subseteq A$ of cardinality p with color n , if A_0 suggests the (p, k_n) -property with respect to $\mathcal{C}_n$ for some n . If we apply Theorem 3.29 with $l := N(D+1)$, $c := d$ and $q := p$, we obtain that there is $P := n_0$ big enough such that every $A' \subseteq A$ with $|A'| = P$ has a one-colored $A'' \subseteq A'$ with $|A''| = N(D+1)$. Since A'' satisfies the (p, k_i) -property for some i by construction of the multigraph, we now know that there are $N_i \leq N$ concepts $\tilde{C} := \{\tilde{c}_1, \dots, \tilde{c}_{N_i}\} \subseteq \mathcal{C}_i$ which cover A'' . There must be some $c'' \in \tilde{C}$ with $|c''| \geq D+1$, since otherwise we would obtain

$$N(D+1) = |A''| \leq \sum_{j=1}^{N_i} |\tilde{c}_j| < N_i(D+1) \leq N(D+1)$$

which obviously cannot hold. Hence we get that for an arbitrary $A' \subseteq A$ with cardinality P there is a $c'' \in \bigcup_{i=1}^d \mathcal{C}_i$ such that

$$D+1 \leq |c'' \cap A''| \leq |c'' \cap A'|,$$

which yields us the $(P, D+1)$ -property. $\square$

Before assembling the final compression scheme, we state two technical lemmas.

Lemma 3.55 *Let $(X, \mathcal{C})$ be a concept class of VC-dimension d , $m \in \mathbb{N}$, $A \subseteq X$, $x \in A^m$. Then the three following statements hold:* 3.

- (i) *If $\text{alt}(x; \mathcal{C}) = 2d+2$, then the subsequence x' of x , which contains all label changes made by some $C \in \mathcal{C}$ (by "contains all label changes we mean that we obtain x' from x by removing all elements which do not contribute to the maximization of alt ") which maximizes alt has even cardinality.*
- (ii) *If m is an even number and $m = \text{alt}(x; \mathcal{C})$, then for each cyclic permutation x_{cycle} of x we get $\text{alt}(x; \mathcal{C}) = \text{alt}(x_{\text{cycle}}; \mathcal{C})$*

We will from now on say that the *alternating* subsequence x_{alt} of a sequence x is the sequence that defines $\text{alt}(x; \mathcal{C})$.

Proof. We will prove the two statements of the lemma individually:

(i): Assume $\text{alt}(x; \mathcal{C}) = 2d + 2$, consider x_{alt} . By Definition 3.35, we obtain that

$$2d + 2 = \text{alt}(x_{\text{alt}}; \mathcal{C}) = \max_{C \in \mathcal{C}} |\{i : C(x_{\text{alt}_i}) \neq C(x_{\text{alt}_{i-1}})\}| + 1 = (|x_{\text{alt}}| - 1) + 1,$$

because of how we defined x_{alt} and that the only i which does not satisfy $C(x_{\text{alt}_i}) \neq C(x_{\text{alt}_{i-1}})$ is $i = 1$; it has no predecessor.

(ii): If m is an even number and $m = \text{alt}(x; \mathcal{C})$, then there is a $C \in \mathcal{C}$, which labels x on the sequence $a = (0, 1, 0, \dots, 1, 0, 1)$ or $b = (1, 0, 1, \dots, 0, 1, 0)$ of size m . Note that $C(x_1) \neq C(x_m)$, since m is an even number. Thus for any cyclic permutation and the resulting sequence x_{cycle} , we get that $C(x) = a$ if and only if $C(x_{\text{cycle}}) = b$, so we obtain that $\text{alt}(x; \mathcal{C}) = \text{alt}(x_{\text{cycle}}; \mathcal{C})$.

□

Lemma 3.56 For every $\vartheta_1, \vartheta_2, \dots, \vartheta_{d+1}$ there exists a p such that: Given 2.
($X, \mathcal{C}$), a concept class of VC-dimension d , for any finite set A and concept C , the set $A \cap C$ has the $(p, \vartheta_1, \vartheta_2, \dots, \vartheta_{d+1})$ property with respect to the concept classes

$$(X, \mathcal{C}^*) = \left(X, \mathcal{AL}\mathcal{T}_{2d+2}(A, C; \mathcal{C}), \mathcal{AL}\mathcal{T}_{\vartheta_1+2d+2}(A, C; \mathcal{C}), \dots, \mathcal{AL}\mathcal{T}_{\sum_{i=1}^d \vartheta_i+2d+2}(A, C; \mathcal{C}) \right).$$

Remark 3.57 Note that this p is not dependent on A or C , but only on 3.
the VC-dimension d .

Proof. Choose p such that any sequence x of size p contains a subsequence y of length $m = \sum_{i=1}^{d+1} \vartheta_i + 2d + 2$ with $\text{alt}(y; C) \leq 2d + 1$ (such a combination of variables exists by Corollary 3.37). We claim that this p will suffice.

Indeed, given a set $A_0 \subseteq A \cap C$ of size p , let t be the maximal number (which may be 0) for which there is a sequence

$$y^t = (y_1, y_2, \dots, y_{\vartheta_{d+1}-t+1}, z_1, y_{\vartheta_{d+1}-t+2}, z_2, \dots, y_{\vartheta_{d+1}-t+t}, z_t, y_{\vartheta_{d+1}-t+t+1}, y_{\vartheta_{d+1}-t+t+2}, \dots, y_n)$$

such that:

1. $\{y_1, y_2, \dots, y_n\} \subseteq A_0$;
2. $\{z_1, \dots, z_t\}$ are in $A \setminus C$;
3. $\text{alt}(y^t; C) \leq 2d + 1$;
4. $|y^t| = \sum_{i=1}^{d+1-t} \vartheta_i + 2d + 2$ (where we define $\sum_{i=1}^{d+1-t} \vartheta_i = 0$ if the upper sum becomes negative).

Note that t refers to the number of elements z in our sequence y .

We now obtain the following properties for t :

- i) $t \geq 0$; Define $M := \{x \in A_0^p\}$ and $M' := \{y: y \text{ is a subsequence with size } m = \sum_{i=1}^{d+1} \vartheta_i + 2d + 2 \text{ and } \text{alt}(y; C) \leq 2d + 1 \text{ of an element } x \in M\}$. Note that $M' \neq \emptyset$, by our assumptions. We now prove that any $y \in M'$ satisfies all conditions and can be selected as y^0 . Hence, $t \geq 0$, because t shall be the maximal number. 1, 3, 4 are all satisfied by construction. 2 is satisfied, because we do not have such elements in y , since $A_0 \subseteq A \cap C$.
- ii) $t < d + 1$; Assume that without loss of generality there is a maximal $t = d + 1$. We would have $d + 1$ elements in our sequence y^t that are in $A \setminus C$ by 2 (those are labeled as 0 by C), which are in between elements of A_0 by 1 (those are labeled as 1 by C). Thus C would cause $2(d + 1) = 2d + 2$ label changes on y^t and as an immediate consequence we would know that $\text{alt}(y; C) \geq 2d + 2$ (by Definition 3.35), which would contradict 3.

Consider the sequence y^{t+1} defined by removing the first ϑ_{d+1-t} elements in y^t . The cardinality of y^{t+1} is $\sum_{i=1}^{d-t} \vartheta_i + 2d + 2$, because

$$|y^{t+1}| = |y^t| - \vartheta_{d+1-t} = \left(\sum_{i=1}^{d+1-t} \vartheta_i \right) - \vartheta_{d+1-t} + 2d + 2 = \left(\sum_{i=1}^{d-t} \vartheta_i \right) + 2d + 2. \quad (3.2)$$

We will now prove through contradiction, that $\text{ALT}(y^{t+1}; C)$ has the honesty condition as in Definition 3.43 with respect to C , because we will need this statement later.

Assume that $\text{ALT}(y^{t+1}; C)$ does not have the honesty condition with respect to C . Then there must be some $z_0 \in A$ such that $z_0 \notin C$ and $\text{alt}((z_0, y^{t+1}); C) \leq 2d + 1$ by Definition 3.38. Based on our assumptions, we will see that we are able to construct a new sequence increasing t , hence breaking the maximality of t through the following scheme:

consider y^{t+1} .

Step 1: Define the sequence $y' := (z_0, y^{t+1})$. We obtain

$$|y'| = \left(\sum_{i=1}^{d-t} \vartheta_i \right) + 2d + 2 + 1,$$

by equation 3.2. consider the subsequence y'' by removing the last $\vartheta_{d-t} + 1$ elements from y' . Like above, we obtain

$$|y''| = \left(\sum_{i=1}^{d-t-1} \vartheta_i \right) + 2d + 2$$

By ii), we know that $t \leq d$, which implies that

$$|y''| \leq \left(\sum_{i=1}^{d-t-1} \vartheta_i \right) + 2t + 2 \quad (3.3)$$

We said earlier, that if the upper sum becomes negative (which can happen, if $t = d$ before we considered y^{t+1}). This and the equation 3.3 guarantees us, that even if $t = d$, we did not touch a single z in y' by removing the last $\vartheta_{d-t} + 1$ elements and the last element is definitely a y -element (because y' begins with a z -element and we get alternately z -and y -elements up until $y_{\vartheta_{d+1-t}+t+1}$; the alternation has $2(t+1) = 2t+2$ elements). So y'' looks like this:

$$y'' = (z_0, y_{\vartheta_{d+1-t}+1}, z_1, y_{\vartheta_{d+1-t}+2}, z_2, \dots, y_{\vartheta_{d+1-t}+t}, z_t, y_{\vartheta_{d+1-t}+t+1}, y_{\vartheta_{d+1-t}+t+2}, \dots, y_{n-\vartheta_{d-t}+1})$$

Step 2: Apply a cyclic permutation that takes the last $\vartheta_{d-t} + 1$ in the sequence y' and moves them to the beginning of the sequence. Define the resulting sequence y'_{new} .

Step 3: We define y''_{new} , by deleting the last element of the sequence y'_{new} . y''_{new} then looks like this:

$$y''_{new} = (y_{n-\vartheta_{d-t}}, \dots, y_n, z_0, y_{\vartheta_{d+1-t}+1}, z_1, \dots, y_{\vartheta_{d+1-t}+t}, z_t, \dots, y_{n-\vartheta_{d-t}-2})$$

We will now prove that y''_{new} satisfies all conditions for y^t .

Condition 1: Since we did not add any new elements while constructing y''_{new} from y^t and by assumption y^t satisfied 1, y''_{new} satisfies 1.

Condition 2: Same argumentation as above; so 2 is satisfied.

Condition 3: We know that $\text{alt}(y^t; C) \leq 2d + 1$, by assumption. First, we removed elements from the sequence y^t , which obviously can not increase $\text{alt}(y^t; C)$. Based on our assumption on z_0 , we also could not have increased $\text{alt}(y^t; C)$, when we defined y' . Assume that we increased alt in step 2, while doing the cyclic permutation. Without loss of generality, we can assume that $\text{alt}(y'_{new}; C) = 2d + 2$, since if it is even higher, we just delete elements which contribute to $\text{alt}(y'_{new}; C)$ and we get the same argumentation. It is sufficient to observe only the alternating subsequence $y_{alternating}$ of y_{new} , which has even cardinality by (i). By (ii), we know that the cyclic permutation of $y_{alternating}$ does not change alt . So what we just did is to remove the "unimportant" elements of y_{new} , so only consider those elements from y_{new} , which contribute to $\text{alt}(y_{new}; C)$. We end up seeing that this number could not have changed (since the alternating subsequence has even cardinality) from $\text{alt}(y'; C)$ which yields us

$$2d + 2 = \text{alt}(y_{new}; C) = \text{alt}(y'; C),$$

which of course is impossible by assumption. So our cyclic permutation did not increase alt and after that we removed an element, which also could not have increased alt . Thus we get $\text{alt}(y''_{new}; C) \leq 2d + 1$

Condition 4: We see

$$|y''_{new}| = |y'| - 1 = \left(\sum_{i=1}^{d-t} \vartheta_i\right) + 2d + 2,$$

because we permuted in step 2, which of course does not change the cardinality and in step 3 we deleted one element which was one too much. So y''_{new} satisfies 4.

Since we end up with one more z -element in y''_{new} , by manipulating y^t , t can not be maximal assuming $\text{ALT}(y^{t+1}; \mathcal{C})$ does not have the honesty condition with respect to C , which is the contradiction we needed. Hence $\text{ALT}(y^{t+1}; \mathcal{C})$ has the honesty condition with respect to C , which means by definition that we have $\text{ALT}(y^{t+1}) \in \mathcal{ALT}_{\sum_{i=1}^{d-t} \vartheta_i + 2d + 2}(A, C; \mathcal{C})$ for some t in the earlier discussed boundaries. Note that $\{y_1, \dots, y_{\vartheta_{d+1-t}}\} \subseteq A_0 \cap \text{ALT}(y^{t+1}; \mathcal{C})$ (by 1 and 3 applied on y^t).

To sum that all up, if we fix an arbitrary set $A_0 \subseteq A \cap C$ of size p , there is a $0 \leq t \leq d$, such that

$$\vartheta_{d+1-t} = |\{y_1, \dots, y_{\vartheta_{d+1-t}}\}| \leq |A_0 \cap \text{ALT}(y^{t+1}; \mathcal{C})|,$$

which means that A_0 suggests the property (p, ϑ_{d+1-t}) with respect to $\mathcal{ALT}_{\sum_{i=1}^{d-t} \vartheta_i + 2d + 2}(A, C; \mathcal{C})$ for some t . Thus, we have proven the Lemma. $\square$

With all combinatorial building blocks in place, we now provide the complete proof of Theorem 3.52 by constructing an explicit, honest compression scheme.

Proof of Theorem 3.52. Let C and $S_C(x) \in S(X^m)$ be given. Because of the equivalence of definitions 3.19 and 3.23, it is sufficient to prove the existence of a word compression scheme with extra bit defined in Definition 3.26. The existence of a compression scheme with additional information then follows as discussed in 3.27

First, let $\tilde{S}$ be the set of all elements $x \in X$ that appear in the sample $S_C(x)$. The VC-dimension of $\mathcal{ALT}_k(\tilde{S}, C; \mathcal{C})$ is bounded by the VC-dimension of $\mathcal{ALT}_k(X; \mathcal{C})$ which is finite by Lemma 3.42.

Define $\vartheta_1 := \text{VC-dim}(\mathcal{ALT}_{2d+2}(X; \mathcal{C})) + 1$. Define inductively for every $n \leq d + 1$:

$$\vartheta_n := \text{VC-dim}\left(\mathcal{ALT}_{\sum_{i=1}^{n-1} \vartheta_i + 2d + 2}(X; \mathcal{C})\right) + 1.$$

Lemma 3.56 states that for an appropriate p , $\tilde{S} \cap C$ has the $(p, \vartheta_1, \vartheta_2, \dots, \vartheta_{d+1})$ property with respect to the $d + 1$ concept classes with common domain X :

$$\left(X, \mathcal{ALT}_{2d+2}(\tilde{S}, C; \mathcal{C}), \dots, \mathcal{ALT}_{\sum_{i=1}^d \vartheta_i + 2d + 2}(\tilde{S}, C; \mathcal{C})\right). \quad (3.4)$$

First, for simplicity, assume that all concepts $\mathcal{ALT}(\tilde{S}, C; \mathcal{C})$ that appear in Equation 3.4 are non-degenerate.

By Theorem 3.54 there are N honest concepts for $\tilde{S}$ with respect to C in $\bigcup \mathcal{ALT}(\tilde{S}, C; \mathcal{C})$, say

$$\{ALT(x_1; \mathcal{C}), \dots, ALT(x_N; \mathcal{C})\},$$

such that for every $x \in \tilde{S} \cap C$, we have $x \in ALT(x_i; \mathcal{C})$ for some i . Each x_i is at most of size $\sum_{i=1}^d \vartheta_i + 2d + 2$. Define:

$$\kappa(S_C(x)) = (x_1, b, x_2, b, \dots, b, x_N).$$

The reconstruction function works as follows: for any $x \in \tilde{S} \cap C$ we put $\rho(x) = 1$ if for some sequence x_i we have $x \in ALT(x_i; \mathcal{C})$; otherwise we set $\rho(x) = 0$. The honesty property ensures that if $C(x) = 0$ then $\rho(x) = 0$. The reason for that is if $C(x) = 0$, then $x \notin C$, because we said that we associate C with its indicator function. The honesty property now says, that if $x \notin C$, then none of the honest $ALT(x_i; \mathcal{C})$ is allowed to have x in it. Hence we obtain $\rho(x) = 0$.

Theorem 3.54 ensures that if $C(x) = 1$, we will have $\rho(x) = 1$. The reason for that we know that each x is covered by $\bigcup_{i=1}^N ALT(x_i; \mathcal{C})$. So if $C(x) = 1$, we obtain $x \in C$ which means that x there is an i such that $x \in ALT(x_i; \mathcal{C})$. So by definition we get $\rho(x) = 1$.

Because each x_i has a bounded size, we know that this compression is bounded by $N \times (\sum_{i=1}^d \vartheta_i + 2d + 2) + N$ for some sequence x , where N is dependent on p , which in turn only depends on d , because of the remark under Theorem 3.56. So the bound we obtain is independent of the size of x , but only on the VC-dimension.

Thus we have constructed bounded a compression scheme with extra bit according to Definition 3.26, if all $\mathcal{ALT}(\tilde{S}, C; \mathcal{C})$ that appear in Equation 3.4 are non-degenerate.

If some of the concepts in Equation 3.4 are degenerate, simply add to the degenerate concept classes $\mathcal{ALT}(\tilde{S}, C; \mathcal{C})$ all the singletons $x_{single} \in \tilde{S} \cap C$, that are not covered yet by $\mathcal{ALT}(\tilde{S}, C; \mathcal{C})$. So we "fill" each of those degenerate concept classes until none of them is degenerate, allowing us to apply the theorems just as in the non-degenerate case. This will increase the VC-dimension by only a constant factor, since $\tilde{S}$ is finite by definition. By the same method of proof one will still obtain for some N , N bounded concepts, $\{\{x_1\}, \{x_2\}, \dots, \{x_M\}, ALT(x_{M+1}; \mathcal{C}), \dots, ALT(x_N; \mathcal{C})\}$. The concepts of the form $ALT(x; \mathcal{C})$ will be honest. As for the singleton concepts, we may assume $\{x_i\}$ is contained in $\tilde{S} \cap C$ since those concepts were used to cover $\tilde{S} \cap C$ (it is possible however, that those concepts are not necessarily needed to actually cover $\tilde{S} \cap C$; we can skip those since our compression scheme must guess only our sample set $\tilde{S}$ correctly) and we add them to the compressed sequence. $\square$

Remark 3.58 Interestingly, the constructed compression scheme is not "just" ^{3.} an ordinary compression scheme according to Definition 3.19, but also an

honest compression scheme, because by construction we had "the honesty property ensures that if $C(x) = 0$ then $\rho(x) = 0$ ".

So given a concept class $(X, \mathcal{C})$ with $\text{VC-dim}(\mathcal{C}) = d$, we have proven that if the alternating concept class $\mathcal{ALT}_{2d+1}(X; \mathcal{C})$ has finite VC-dimension, then we find an honest compression scheme. By Theorem 3.41, we get that if $(X; \mathcal{C})$ has finite VC-dimension, then $\mathcal{ALT}_{2d+1}(X; \mathcal{C})$ has finite VC-dimension, so Theorem 3.52 holds. Thus this assumption is not a restriction at all.

Remark 3.59 One might assume that if the exact structure of an honest k -compression scheme were known, one could bypass training entirely by simply applying the scheme directly over a realizable set. However, computing such compressions is generally *NP-hard*, meaning the computational cost would often exceed that of standard machine learning algorithms. Therefore, the true strength of an existence statement is not algorithmic substitution, but the provision of a guaranteed sample-size bound. Knowing that k consistent examples suffice to prevent overfitting provides a strong theoretical foundation: it assures us that if a model—such as a neural network—is trained incrementally (meaning., via batch training with algorithms like *Adam*), k consistent examples are inherently sufficient to learn a robust concept, making the training process theoretically sound from a data-quantity perspective. Thus we might ask ourselves if the bound established in Theorem 3.52 might be improved. 3.

Example 3.60 Since $(\mathbb{R}, \mathcal{C}_{tf})$ and $(\mathbb{R}, \mathcal{C}_{hs})$ have finite VC-dimension, they admit bounded, honest sample compression schemes by Theorem 3.52. 3.

3.7 Further analysis of the obtained boundaries for compression schemes

Having established the existence of honest compression schemes in Theorem 3.52, a natural question arises regarding the quantitative efficiency of this construction. Specifically, the bound $O(\sum_{i=1}^d \vartheta_i + 2d + 2)$ derived via alternating concept classes can be extremely large due to the iterated tower-like growth of the ϑ_i terms. To analyze whether this bound can be improved, we briefly detour into the theory of dual concept classes, where the roles of domain points and concepts are swapped.

Definition 3.61 (Dual concept class [MY15]) Let $(X, \mathcal{C})$ be a concept class. 0. We define the *dual concept class* as the concept class $(X^*, \mathcal{C}^*)$, where the domain is $X^* = \mathcal{C}$ and the family of concepts is $\mathcal{C}^* = \{C_x : x \in X\}$ with

$$C_x = \{C \in \mathcal{C} : x \in C\}.$$

Example 3.62 The dual concept class of half-spaces $(\mathbb{R}^d, \mathcal{C}_{hs})^*$ is given by 3.

$$(\mathcal{C}_{hs}, \mathcal{C}_{hs}^*), \quad \text{where} \quad \mathcal{C}_{hs}^* := \{C_x \subseteq \mathcal{C}_{hs} : x \in \mathbb{R}^d\}$$

and for each $x \in \mathbb{R}^d$,

$$C_x = \{y \in \mathbb{R}^d : w \cdot y + b \leq 0\} \in \mathcal{C}_{hs} : w \cdot x + b \leq 0\}.$$

Equivalently, each instance C_x consists of all half-spaces that contain the point x .

Definition 3.63 (Dual shattering and dual VC-dimension) Given a concept class $(X, \mathcal{C})$ and its dual concept class $(X^*, \mathcal{C}^*)$, a finite set of concepts $A \subseteq \mathcal{C}$ is said to be *dually shattered* if

$$\{A \cap C_x : x \in X\} = 2^A = \mathcal{P}(A).$$

2. Synthesized from multiple sources, independently worded.

The *dual VC-dimension* of the concept class, $\text{VC-dim}^*(\mathcal{C})$, is defined as the maximal cardinality of a finite set of concepts $A \subseteq \mathcal{C}$ that is dually shattered. If such a maximum doesn't exist, we write $\text{VC-dim}^*(\mathcal{C}) = \infty$.

Lemma 3.64 ([Sim15]) For any concept class $(X, \mathcal{C})$, we have

1.

$$\text{VC-dim}^*(\mathcal{C}) < 2^{\text{VC-dim}(\mathcal{C})+1} \quad \text{and} \quad \text{VC-dim}(\mathcal{C}) < 2^{\text{VC-dim}^*(\mathcal{C})+1}.$$

In particular, $\mathcal{C}$ has finite VC-dimension if and only if it has finite dual VC-dimension.

Proof. We show $\text{VC-dim}(\mathcal{C}) < 2^{\text{VC-dim}^*(\mathcal{C})+1}$; the dual bound follows symmetrically.

Assume $\text{VC-dim}(\mathcal{C}) \geq 2^n$ for some $n \in \mathbb{N}$. Then there exists a set $A \subseteq \mathcal{C}$ with $|A| = 2^n$ that is shattered by $\mathcal{C}$. Label the elements of A as $A = \{a_S : S \subseteq \{1, \dots, n\}\}$ (so $|A| = 2^n$). For each $k \leq n$, let $C_k \in \mathcal{C}$ be such that:

$$C_k \cap A = \{a_S : k \in S \subseteq \{1, \dots, n\}\}.$$

Now consider the family $\{C_1, \dots, C_n\} \subseteq \mathcal{C}$. For any subset $I \subseteq \{1, \dots, n\}$, define the dual point:

$$x_I := \{C \in \mathcal{C} : a_I \in C\} \in X^*.$$

The key observation is that for any $I \subseteq \{1, \dots, n\}$, the set $\{C_k : k \in I\}$ is exactly the set of concepts in $\{C_1, \dots, C_n\}$ containing x_I . Thus, the family $\{C_1, \dots, C_n\}$ is shattered by $\mathcal{C}^*$, which implies:

$$\text{VC-dim}^*(\mathcal{C}) \geq n.$$

Since this holds for arbitrary n , we conclude that:

$$\text{VC-dim}(\mathcal{C}) < 2^{\text{VC-dim}^*(\mathcal{C})+1}.$$

The first inequality follows symmetrically. □

Example 3.65 By Example 3.8 and Lemma 3.64, we know that $\text{VC-dim}^*(\mathcal{C}_{hs})$ is bounded by 2^{d+2} . 3.

The equivalence between finite VC-dimension and finite dual VC-dimension allows us to apply a result by Moran and Yehudayoff, which drastically improves the size bound for general sample compression schemes.

Theorem 3.66 (Compression using dual VC-dimension [MY15]) *If a concept class $(X, \mathcal{C})$ has VC-dimension $d > 0$ and its dual concept class $(X^*, \mathcal{C}^*)$ has dual VC-dimension $d^* > 0$, then $\mathcal{C}$ has a sample compression scheme of size $k \log k$ with $k = O(d^* \cdot d)$.* 0.

Combining Lemma 3.64 with Theorem 3.66 immediately yields a much tighter, purely combinatorial size bound for sample compression. However, as we discuss below, this reduction comes at a significant conceptual cost regarding model-theoretic applications.

Theorem 3.67 *If a concept class $(X, \mathcal{C})$ has finite VC-dimension, then $\mathcal{C}$ has a bounded sample compression scheme.* 3.

While the result of Moran and Yehudayoff 3.66 guarantees a dramatically tighter bound than in Theorem 3.52, their construction relies on min-max arguments that do not preserve the honesty condition. For model-theoretical applications—specifically the construction of UDTFS and parameter-efficient definable reconstructions—honesty is strictly required. Consequently, despite its large theoretical bound, the Livni-Simon construction remains indispensable in our framework.

4 Formalization of our main result

Having established the combinatorial machinery of honest compression schemes in the previous Chapter, we now bridge these results back to model theory. We formalize that UDTFS is the precise model-theoretic counterpart to honest sample compression. Moreover, we can finally prove that a formula being NIP is equivalent to it satisfying UDTFS.

We do important observation that UDTFS is the model-theoretical analogue to honest compression schemes, discussed in chapter 3. UDTFS should be thought of as saying that there is a definable compression scheme for the family $\mathcal{F}_\varphi$. The compression function takes as input a finite subset A and the trace $\varphi(A; a)$ and outputs the tuple b . The reconstruction function takes the tuple b as input and outputs the definable set $\psi(A; b)$.

Theorem 4.1 (Translation of compression schemes) *Some concept class $(X, \mathcal{C})$ has an honest k -compression scheme if and only if the corresponding formula in the corresponding $\mathcal{L}$ -structure has UDTFS.* 3.

Proof. Let $(X, \mathcal{C})$ have a k -compression scheme consisting of a compression function κ and a reconstruction function ρ according to Definition 3.19. Let $\mathcal{M}$ be the corresponding $\mathcal{L}$ -structure and $\varphi(x; y)$ the $\mathcal{L}$ -formula defining the family $\mathcal{F}_\varphi = \mathcal{C}$.

Since $\rho: S(X^k) \times X \rightarrow \{0, 1\}$ determines whether an element $x \in X$ belongs to the reconstructed concept based on a labeled k -sample, there exists an $\mathcal{L}$ -formula $\psi(x; z_1, \dots, z_k)$ representing this reconstruction rule. We globally define our UDTFS candidate formula as:

$$\vartheta(x; z_1, \dots, z_k) := \psi(x; z_1, \dots, z_k).$$

Now, fix a finite subset $A \subseteq X$ and a parameter $a \in M$. Let $C = \varphi(M; a) \in \mathcal{C}$ be the corresponding concept.

Applying the compression function κ to the labeled sample $S_C(A)$ yields a subsequence of $S_C(A)$ of length k . The domain elements of this compressed sample form a tuple $b = (b_1, \dots, b_k) \in A^k$.

By Condition 2 of Definition 3.19, the reconstruction function ρ satisfies:

$$\rho(\kappa(S_C(A)), x) = 1 \iff x \in C \quad \text{for all } x \in A.$$

By construction of ϑ and b , this condition is directly equivalent to the model-theoretic statement:

$$\mathcal{M} \models \vartheta(x; b) \iff \mathcal{M} \models \varphi(x; a) \quad \text{for all } x \in A.$$

Hence, we have $\varphi(A; a) = \vartheta(A; b)$ with $b \in A^k$. Since the formula $\vartheta(x; z)$ was defined independently of A and a , this demonstrates that $\varphi(x; y)$ has UDTFS. $\square$

Combining our main learning-theoretic result Theorem 3.52 with the translation principle Theorem 4.1, we obtain the resolution of the generalized UDTFS conjecture for NIP formulas.

Theorem 4.2 (Expanded UDTFS conjecture) *Let M be an $\mathcal{L}$ -structure and $\varphi(x; y)$ some NIP formula.* 3.

- 1) *Then it satisfies UDTFS.*
- 2) *The formula $\vartheta(x; y)$ we obtain from UDTFS satisfies $\vartheta(M; b) \subseteq \varphi(M; a)$.*

Proof of 1. Define the concept class $(M^{|x|}, \mathcal{F}_\varphi)$. By Theorem 3.52, we know that there is some honest k -compression scheme for this concept class, since $\varphi(x; y)$ is NIP and thus the concept class has finite VC-dimension. Theorem 4.1 now says, that $\varphi(x; y)$ satisfies UDTFS. $\square$

Proof of 2. By Theorem 3.52, we know that the compression scheme we obtain is honest. Since $\vartheta(x; y)$ corresponds with the reconstruction function ρ by Theorem 4.1, we know that each $x' \in M^{|x|}$ that is labeled by ρ as "1", must be labeled as "1" by a as well. The model-theoretical translation of this is exactly statement 2. $\square$

To establish the full equivalence between NIP and UDTFS, we recall a characterization by Chernikov and Simon linking definable selections to polynomial density bounds.

Theorem 4.3 ([CS12]) *Let M be an $\mathcal{L}$ -structure and $\varphi(x; y)$ some $\mathcal{L}$ -formula, then the following are equivalent:* 0.

1. *There is a formula $\vartheta(x; z)$ such that for any set A and any parameter a , there exists $b \in A^k$ satisfying $\varphi(A, a) = \vartheta(A, b)$.*
2. *There exist $m, n \in \mathbb{N}$ such that $|S_\varphi(A)| \leq m \cdot |A|^n$ for any set A .*

Hence by using Theorem 4.2 and Lemma 2.19 we can finally conclude:

Corollary 4.4 *Let M be an $\mathcal{L}$ -structure and $\varphi(x; y)$ some formula, then the following are equivalent:* 3.

1. *$\varphi(x; y)$ is an NIP formula*
2. *$\varphi(x; y)$ satisfies UDTFS*

Remark 4.5 The corollary yields another option to derive the equivalence 3. between the existence of an honest, bounded sample compression scheme and the finite VC-dimension of a concept class, since we can translate UDTFS into honest, bounded sample compression schemes by Theorem 4.1.

This completes the characterization: a formula $\varphi(x; y)$ is NIP if and only if it admits UDTFS. This result highlights the deep interplay between combinatorial learnability and model theory.

5 Conclusion and Outlook

In this thesis, we explored the fruitful intersection between model theory and statistical learning theory, highlighting how structural concepts in logic parallel generalization bounds in machine learning. By establishing a local, formula-based link between NIP formulas and UDTFS, we utilized the combinatorial machinery of honest sample compression schemes to bridge both domains.

However, several limitations and open avenues for future research remain:

- **Restriction to Binary Classification:** Our analysis focused strictly on binary concept classes and $\mathcal{L}$ -formulas evaluating to truth values in $\{\text{True}, \text{False}\}$, as reflected in the power-set construction in Theorem 3.34. While this boolean framework is sufficient for classical model-theoretical types it restricts the learn-theoretical implications to binary classification tasks. Extending these combinatorial techniques to multiclass classification or continuous regression problems remains an open challenge.
- **Non-Constructive Existence and Computational Complexity:** Our results guarantee the existence of honest compression schemes, but do not provide an explicit, efficient algorithm to construct them. Finding optimal (with respect to the compression size) honest compression schemes is known to be *NP-hard* in general, meaning there is likely no polynomial-time algorithm that can compute such a scheme unless $P = NP$. A promising direction for future work is the design of polynomial-time approximation algorithms or greedy heuristics that yield near-optimal honest compression schemes.
- **Algorithmic Bounds vs. Model-Theoretic Expressiveness:** While general concept classes with finite VC-dimension admit standard compression schemes with a tighter bound as established by Moran and Yehudayoff, the strict requirement of *honesty*—which is essential in model theory—imposes stronger structural constraints. Further investigation is needed to fully characterize the exact trade-offs between the compression size k and the expressiveness of the reconstructing formulas ψ .
- **Extension to the Agnostic Setting:** Our entire analysis, and specifically the existence of honest compression schemes, fundamentally relies on the realizable setting. However, in real-world applications, the agnostic setting is far more prevalent. In practice, data is often subject to noise (measurement errors, statistical inconsistencies et cetera), meaning no single concept $C \in \mathcal{C}$ perfectly labels the training data. The honesty condition inherently requires a definitive ref-

erence C to ensure the reconstructed concept produces no false positives. Consequently, extending honest compression schemes to the agnostic case, while preserving the strict definability required for the model-theoretic translation to UDTFS—remains a highly non-trivial open problem. One approach to bridge this gap could involve the application of *Cover’s theorem* and the *Kernel trick*. Since Cover’s Theorem states that the probability of separability increases in sufficiently higher dimensional spaces, one could potentially lift a noisy, non-separable concept class into a higher-dimensional feature space. By utilizing kernel methods, it might be possible to find a realizable sample within a transformed space, thereby recovering the structural prerequisites needed to construct honest compressions. However extending the dimension will also increase the complexity of the learning problem itself, also known as the *curse of dimensions*.

- **Structural and Syntactic Criteria for UDTFS:** Directly verifying that a structure is NIP can be mathematically challenging. Throughout this work, we highlighted o-minimality. Combined with *quantifier-elimination* as seen in [Hie26] yields a powerful, syntactically accessible strategy to establish NIP. A natural question for future inquiry is whether other structural criteria can provide alternative, easy-to-verify pathways to establish UDTFS directly.
- **UDTFS Beyond First-Order Logic:** While our results operate strictly within classical first-order logic, there are other logical formalisms—such as continuous logic, where metric structures are defined via bounded continuous functions and moduli of uniform continuity. Investigating whether analogues of UDTFS can be formulated in such non-classical frameworks with an equivalence to easy-to-verify properties, presents a compelling direction for future research.

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Deutsche Zusammenfassung / German summary

Die vorliegende Bachelorarbeit untersucht die tiefgehenden strukturellen und funktionellen Verbindungen zwischen der Modelltheorie der mathematischen Logik und der statistischen Lerntheorie. Im Zentrum der Arbeit steht das Zusammenspiel zwischen der Eigenschaft ohne Unabhängigkeit (*No Independence Property*, NIP) sowie der gleichmäßigen Definierbarkeit von Typen über endlichen Mengen (*Uniform Definability of Types over Finite Sets*, UDTFs) auf der modelltheoretischen Seite und der Vapnik-Chervonenkis-Dimension (VC-Dimension) sowie ehrlichen Stichproben-Kompressionsschemata (*honest sample compression schemes*) auf der lerntheoretischen Seite.

Zu Beginn werden die logischen Grundlagen bereitgestellt und die UDTFs-Vermutung im Kontext von NIP-Theorien analysiert. Dabei wird gezeigt, wie sich globale Modell-Eigenschaften durch lokale Typen-Charakterisierungen beschreiben lassen. Anschließend wird der lerntheoretische Rahmen eingeführt, in dem die VC-Dimension als Maß für die kombinatorische Komplexität einer Konzeptklasse dient. Ein wesentliches Hauptresultat der Arbeit ist der detaillierte Beweis des Satzes von Livni und Simon: Konzeptklassen mit endlicher VC-Dimension besitzen stets ein beschränktes, ehrliches Kompressionsschema. Dieses Resultat wird mithilfe von $(d + 1)$ -homogenen und alternierenden Sequenzen sowie einer Verallgemeinerung des (p, k) -Satzes von Matoušek erarbeitet.

Schließlich werden beide Perspektiven zusammengeführt, um zu demonstrieren, wie die Existenz ehrlicher Kompressionsschemata genutzt werden kann, um UDTFs lokal für spezifische Formeln und Typen nachzuweisen. Damit liefert die Arbeit einen anschaulichen interdisziplinären Brückenschlag, der zeigt, wie Methoden der statistischen Lerntheorie zur Lösung modelltheoretischer Fragestellungen beitragen können.